



Targeting drug-resistant tuberculosis: Thioacetamide-substituted benzothiazoles as bactericidal agents inhibiting type II NADH dehydrogenase

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ABSTRACT

The identification of novel anti-tubercular agents that inhibit the function of less-explored yet essential components of *Mycobacterium tuberculosis* is critically needed to address the increasing prevalence of drug-resistant tuberculosis. The type II NADH dehydrogenase, an indispensable component of the oxidative phosphorylation pathway, is the entry point of electrons into the electron transport chain, facilitating ATP production. This enzyme is a promising drug target because its inhibition leads to a bactericidal response through energy starvation. In our study, we synthesized a series of 6-isopropyl benzothiazole derivatives, with various thioacetamide side chains; and assessed their anti-tubercular efficacy, and specificity toward NADH dehydrogenase through *in vitro* studies. The most potent compounds, **C4-1** (MIC = 4 µg/mL, 13.042 µM), and **C4-11** (MIC = 8 µg/mL, 24.80 µM), exhibited inhibitory effects against mycobacterial strains, exhibiting resistance to FDA approved anti-tubercular drugs; and demonstrated bactericidal activity with a safety index exceeding 10. The primary target for these derivatives is proposed to be type II NADH dehydrogenase, indicated by increased NADH/NAD⁺ ratios in the peredox mCherry assay, and from detection of single-nucleotide polymorphisms at multiple points in the type II NADH dehydrogenase gene coding sequence of the strains, exhibiting resistance to the compounds **C4-1**, and **C4-11**. Additionally, molecular docking, molecular dynamics, and *in silico* ADME profiling were performed on the active compounds.

1. Introduction

The bacillus *Mycobacterium tuberculosis* (*Mtb*) is the etiological agent of the debilitating disease tuberculosis (TB) [1]. Despite the availability of multiple drugs for the prevention, and treatment of TB, it remains one of the leading causes of mortality globally. According to the “Global Tuberculosis Report 2024” by the World Health Organization (WHO), TB was the leading cause of death from a single infectious agent in 2023, after marginally trailing behind COVID-19 over the past three years [2]. Although the recovery rate for TB treatment is 85 %, there were still 1.25 million deaths attributed to TB infection in 2023. The rise of drug-resistant TB (DR-TB); which refers to those exhibiting resistance to marketed anti-TB drugs, like: Isoniazid, Rifampicin, Pyrazinamide, Ethambutol, **BDQ** etc.; has significantly contributed to this death toll [2–5].

DR-TB has been classified into five groups: (a) Isoniazid-resistant TB; (b) Rifampicin-resistant TB (RR-TB); (c) Multidrug-resistant TB (MDR-TB), exhibiting resistance to both Isoniazid and Rifampicin; (d) Pre-extensively DR-TB (pre-XDR-TB), where an individual encounters resistance to Rifampicin and a second-line anti-TB drug, Fluoroquinolones [6,7]; and (e) Extensively DR-TB (XDR-TB), showing resistant to either Bedaquiline (**BDQ**) or Linezolid, in addition to being resistant to Rifampicin and Fluoroquinolones [8,9]. In 2023, RR-TB accounted for 400,000 TB cases, raising serious concerns as affected individuals do not respond to Rifampicin, the most effective first-line anti-TB drugs [2]. The rapid emergence of resistance to the newly approved anti-TB drugs, such as **BDQ**, Pretomanid, and Delamanid, have diminished their usefulness in redesigned treatment regimens for treating the DR-TB patients further [8,10,11].

To effectively cure the DR-TB cases, there is an urgent requirement of

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identifying new scaffolds for the development of anti-TB drugs [12–16]. A close insight into the mechanism of drug resistance to the already approved anti-TB drug has emphasized the necessity of designing drugs that remain effective against both replicating and non-replicating strains of *Mtb* [17–19]. *Mtb* may survive under harsh environmental conditions like hypoxia, nutrient scarcity, acidic and basic surroundings, and antibiotic-impregnated environments; by entering into a non-replicating, dormant state. Under such halted growth period of *Mtb*, the operational respiratory chain maintains the minimum energy requirement through the oxidative phosphorylation pathway [20–24]. The approval of **BDQ** for the treatment of MDR-TB, which inhibits the ATP synthase enzyme in the oxidative phosphorylation pathway, and subsequently terminate the ATP production, has encouraged the researchers to target the different components of oxidative phosphorylation pathway for novel anti-TB drug discovery [12,25,26].

Mtb contains two types of NADH dehydrogenase enzymes in the electron transport chain (ETC) of the oxidative phosphorylation pathway: proton pumping type I NADH dehydrogenase (NDH-1) and non-proton pumping type II NADH dehydrogenase (NDH-2). The NDH-2 enzyme conduits electrons into the ETC, which is essential for maintaining the functionality of the ATP synthesis. If NDH-2 remains inactive, other components of the ETC do not transport enough electrons to sustain the ETC operation [27,28]. *Mtb* possesses two sub-types of NDH-2, namely, *ndh* and *ndhA*. Genetic knockout studies have demonstrated the critical role of the *ndh* gene for the survival of *Mtb*. Considering the essentiality of *ndh* for the endurance of *Mtb*, targeting NDH-2 for anti-TB drug development presents a prudent approach [29–34]. While NDH-1 is similar to its human counterpart, found in the mitochondrial ETC; NDH-2 is absent in vertebrate ETC, and is unique to mycobacteria [30,31,35]. Therefore, targeting NDH-2 is a vigilant strategy, as there is no risk of target-specific toxicity in humans, due to the absence of this enzyme in the human ETC.

In 2018, Harbut et al. and Murugesan et al. independently reported the 2-mercaptoquinazolinone scaffold as an NDH-2 inhibitor of *Mtb* (**CBR-5992**, Fig. 1a) [13,36]. Our research group has recently identified two classes of NDH-2 inhibitors: 2-aryl benzimidazole (Fig. 1b) and 2-mercaptobenzothiazole (Fig. 1c) derivatives [37,38]. Our previous work on NDH-2 inhibitors featuring the 2-mercaptobenzothiazole scaffold explored the effects of various substitutions on the benzothiazole core while keeping the cyclohexane moiety in the acetamide side chain intact. Compound **C4** (Fig. 1c) emerged as the most active compound of that series [38]. In our current research, we explored the impact of various substitutions on the side motif of the thioacetamide linker of the benzothiazole acetamide class; while retaining the isopropyl substitution at the C-6 position of the benzothiazole core unaltered (Fig. 1d).

2. Results and discussion

2.1. Chemistry

We synthesized a series of 6-isopropyl-2-mercaptobenzothiazole acetamide derivatives, designated as **C4**, and **C4-1** to **C4-28**, as

outlined in Scheme 1. Intermediate **A**, 6-isopropyl, 2-mercapto benzothiazole was synthesized by reacting 2-bromo-4-isopropyl aniline with potassium ethyl xanthate [39,40]. Intermediate **B** was obtained via the 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDC) coupling reaction between bromoacetic acid, and various alicyclic, alkyl, aryl, heteroaryl, and azacyclic amines [40]. The final products; **C4**, and **C4-1** to **C4-28**, were synthesized via the condensation reaction of the intermediates **A** and **B**, under necessary reaction conditions [41]. Structure of compounds **C4**, **C4-1** to **C4-28** are present in Table 1.

2.2. Minimum inhibitory concentration and structure-activity relationship

All the synthesized 6-isopropyl-2-mercapto benzothiazole acetamide derivatives (**C4**, **C4-1** to **C4-28**) were subjected to the liquid broth microdilution assay, for determining their minimum inhibitory concentration (MIC), against the mycobacterium strain *Mtb* mc²6230 (Table 2, data are expressed as the mean $\pm$ SDs of triplicate). We attempted to note down a structure-activity relationship (SAR) of the synthesized 6-isopropyl-2-mercapto benzothiazole derivatives based on their MIC values. The parent compound **C4** has a MIC of 4 μ g/mL (11.476 μ M), containing cyclohexyl ring substituted acetamide side chain [36]. To explore the effect of varying the ring size of the alicyclic side motif, we synthesized the derivatives **C4-1** to **C4-4**. On reducing the ring size to five (**C4-3**, MIC = 32 μ g/mL, 99.84 μ M), and four carbon atoms (**C4-2**, MIC = 16 μ g/mL, 49.92 μ M), MIC was increased by 8 and 4-folds respectively, than the parent compound **C4** in μ g/mL unit. Further reducing the ring size to three carbon atoms, led to a surprising improvement in MIC value in **C4-1** than the four and five membered ring substituted derivatives (**C4-1**, MIC = 4 μ g/mL, 13.042 μ M), marginally levelling the MIC to that of parent compound **C4**. Activity was deteriorated on increasing the ring size from cyclohexyl (**C4**) to cycloheptyl group (**C4-4**), and for the three cyclohexyl ring fused adamantane ring substituted derivative **C4-5**, causing a 16-folds increase in MIC (in μ g/mL) with respect to **C4**.

Next, we moved to the aliphatic side chains (**C4-6** to **C4-11**). We conceive that for *n*-alkane substituted derivatives, an increase in chain length led to reduction in MIC, as on moving from *n*-propyl to *n*-hexyl substituted derivatives, a gradual reduction in MIC was experienced (**C4-6**, MIC = 64 μ g/mL, 207.48 μ M; **C4-7**, MIC = 32 μ g/mL, 99.22 μ M; **C4-8**, MIC = 16 μ g/mL, 47.54 μ M; **C4-9**, MIC = 16 μ g/mL, 43.88 μ M). For the branch chain substituted derivative **C4-8**, activity was in the same range as of the straight chain alkanes, bearing equal number of carbon atoms (**C4-10**, MIC = 32 μ g/mL, 99.22 μ M). In case of *N*, *N*-diethyl substituted derivative **C4-11**, an improved in MIC value was noted than the mono-alkane substituted derivatives **C4-6** to **C4-10**; leading to a MIC value of 8 μ g/mL (24.80 μ M).

After exploring the effect of aliphatic substitutions, we shifted focus to aromatic substitutions. No improvement in inhibitory response was noted for all the aryl ring substituted derivatives **C4-12** to **C4-18**. All the derivatives have MIC in the range of 64–128 μ g/mL (178.51–343.57 μ M), except **C4-13**. A MIC of 32 μ g/mL (85.90 μ M) was displayed by **C4-13**, bearing the electron-donating methoxy group substituted

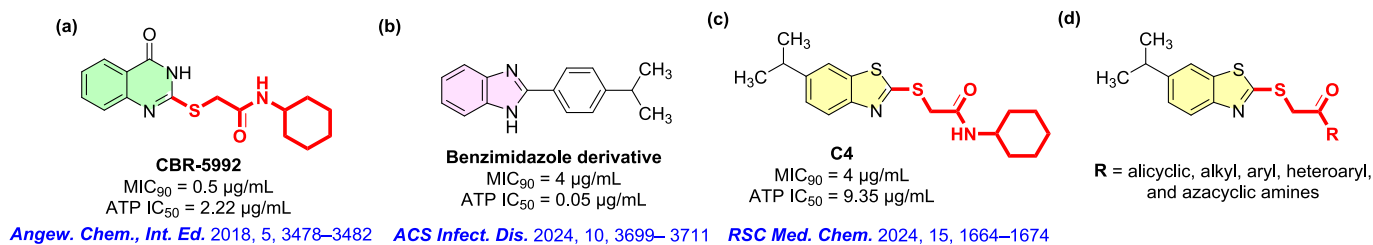
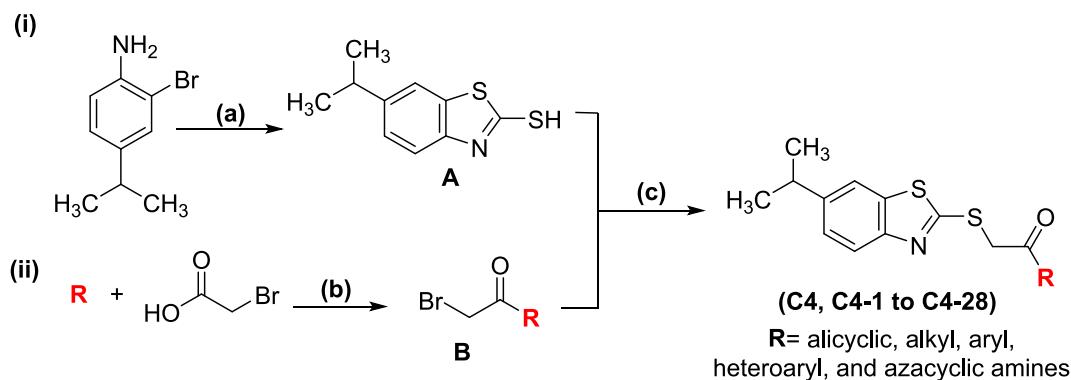


Fig. 1. (a–c) Literature reported NDH-2 inhibitors and their minimum inhibitory concentration (MIC₉₀) and ATP IC₅₀ values against *Mtb* H37Rv mc²6230 strain: 2-mercaptoquinazolinone scaffold containing derivative **CBR-5992**, 2-aryl benzimidazole derivative, and 2-mercaptobenzothiazoles scaffold containing derivative **C4**; (d) General structure of the 6-isopropyl, 2-mercaptobenzothiazole derivatives under investigation.



Scheme 1. Synthetic scheme of 2-(6-isopropylbenzo[d]thiazole-2-ylthio)-N-acetamide derivatives^a.

^a**Reagents and conditions:** (a) 2-bromo-4-isopropylaniline (1 mmol), potassium ethyl xanthate (1.5 equiv.), DMF, 165 °C, 6 h; (b) Alicyclic, alkyl, aromatic, heteroaromatic and azacyclic amines (1 equiv.), bromoacetic acid (1 equiv.), EDC. HCl (1 equiv.), DCM, room temperature, 12 h; (c) **A** (1 equiv.), **B** (1 equiv.), DMF, DIPEA (1 equiv.), room temperature, 2–3 h.

derivative on the phenyl ring.

Next, we shifted to exploring the effect of introducing heteroaromatic ring systems via the amide linkage (**C4-19** to **C4-25**). All the heteroaromatic ring system substituted derivatives possess poor MIC; in the range of 64–128 µg/mL (162.63–366.04 µM). Lastly, the effect of introducing nitrogen containing, heterocyclic ring systems were explored. A MIC of 64 µg/mL was exhibited by the piperidine, and morpholine ring substituted derivatives; **C4-26** (191.33 µM), and **C4-27** (190.19 µM), respectively. For the thiomorpholine ring bearing derivative, a 1-fold reduction in MIC (**C4-28**, MIC = 32 µg/mL, 90.77 µM) was witnessed than the rest of the nitrogen containing, heteroaryl ring system bearing derivatives.

From the observations of the SAR of the 6-isopropyl-2-mercapto benzothiazole acetamide derivatives (**C4**, **C4-1** to **C4-28**) we may infer that alicyclic and alkyl substitutions are favoured at the acetamide linker over aromatic, heteroaromatic, and azacyclic substitutions. Among the alicyclic and alkyl substituted derivatives, alicyclic ring substitution with a specific ring size is essential for activity. We assume ring strain and lipophilicity profile together have some role to play at this point, from the MIC trend. Overall, an optimum lipophilicity profile, without introducing additional steric factor is necessary for securing the desired inhibitory response. Out of the 28 synthesized 2-mercapto benzothiazole acetamide derivatives (**C4-1** to **C4-28**), cyclopropyl ring substituted derivative **C4-1** and *N,N*-diethyl substituted **C4-11** were identified as the active molecules with MIC in the range of 4–8 µg/mL. The compound with the lowest MIC value of the series i.e., **C4-1** (MIC = 4 µg/mL, 13.042 µM), have displayed marginally high MIC value than the parent compound **C4** (MIC = 4 µg/mL, 11.476 µM). This observation directed us to the conclusion that, cyclohexyl ring is the most preferred substitution in the acetamide side chain; and introduction of any alteration at this acetamide site leads to detrimental response.

2.3. Intracellular ATP depletion assay

The oxidative phosphorylation pathway plays an imperative share in the survival of *Mtb*, by serving the role of energy-producing machinery, in terms of ATP. Hampering the operation of any essential components of the oxidative phosphorylation pathway, therefore leads to the interruption of the ATP synthesis, and is reflected by a depleted intracellular ATP level. To assess the effect of our synthesized derivatives (**C4**, **C4-1** to **C4-28**) on the performance of the oxidative phosphorylation pathway, an intracellular ATP-depletion assay was conducted on aerobically grown *mc*²6230. The ATP IC₅₀ values of all the derivatives are presented in Table 2. The same trend of MIC values was observed in ATP IC₅₀ values. Notably, **C4-1**, and **C4-11**, the two derivatives that show the lowest MIC values in the series, have exhibited minimum ATP IC₅₀

values as well. Additionally, we were pleased to find that, the ATP IC₅₀ value of **C4-1**, and **C4-11** were lower than the parent compound **C4**; though **C4**, and **C4-1** possess MIC in similar range.

When the test active compounds, namely, **C4-1**, and **C4-11** were studied for ATP depletion potential, against nutrient starved (NS), non-replicating strain of *mc*²6230; pleasingly a much potent inhibitory response was noted; indicated by the improved ATP IC₅₀ values of 5.99 ± 1.45 µg/mL (19.56 ± 1.45 µM), and 4.91 ± 1.83 µg/mL (15.22 ± 1.83 µM) respectively, by both the derivatives (**C4-1**, and **C4-11**) (Table 3, Fig. 2). We see, despite of having higher MIC and ATP IC₅₀ values against *Mtb mc*²6230; **C4-11** displayed lower ATP IC₅₀ value against NS strain, than **C4-1**. BDQ, and literature reported NDH-2 inhibitor **CBR-5992** were employed as positive control in the experiment. We infer; such response is an indication of the inhibitory potential of our compounds under investigation against the non-replicating, and dormant state of *Mtb*.

2.4. Potency against drug-resistant *Mtb* strains

Identification of anti-TB compounds, that are equally effective against mono, and multi-drug resistant *Mtb* strains are severely needed to address the challenges posed by the ineffectiveness of existing anti-TB drugs in treating patients with DR-TB. Our test active compounds, **C4-1** and **C4-11** were screened against the following strains: *mc*²8243 (Isoniazid-resistant strain), *mc*²8247 (Rifampicin resistant strain), and multidrug-resistant strain: *mc*²8259 (Isoniazid + Rifampicin resistant strain). Based on the MIC values mentioned in Table 4, we may infer that both compounds demonstrate comparable potency against drug resistant *Mtb* strains. **C4-1** and **C4-11** have comparable potency against the RR strain and MDR strain as well.

2.5. Minimum bactericidal concentration determination

All the anti-mycobacterial agents are preferred to be bactericidal because a bacteriostatic response only suppresses growth and reproduction, without ensuring the complete eradication of existing viable bacterial populations. In our study, we investigated two active compounds, **C4-1** and **C4-11**, to determine whether they are bactericidal or bacteriostatic. For that, *Mtb mc*²6230 strain was exposed to different concentrations in multiples of the MIC of **C4-1** and **C4-11** for five, and ten days. It was observed that, both the tested molecules are bactericidal, significantly reducing the bacterial load at 4 × MIC on fifth day and at 2 × MIC on tenth day (Fig. 3A, B). When the test active compounds were subjected to MBC determination against nutrient starved *mc*² 6230 strain, bactericidal response was noted; with significant reduction in the bacterial load at 4 × MIC on 15th day (Fig. 4A, B).

Table 1

Structure of the synthesized tioacetamide-substituted 6-isopropyl, benzothiazole derivatives, with defined substitutions as “R”:

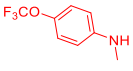
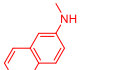
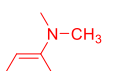
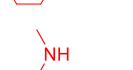
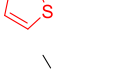
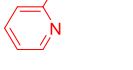
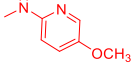
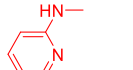
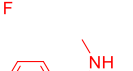
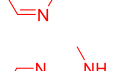
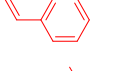
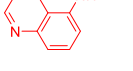
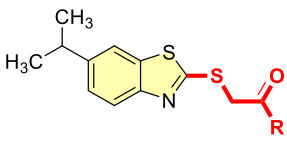
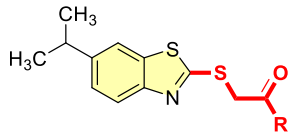
Serial no.	Compound code	R	Serial no.	Compound code	R
1	C4		17	C4-16	
2	C4-1		18	C4-17	
3	C4-2		19	C4-18	
4	C4-3		20	C4-19	
5	C4-4		21	C4-20	
6	C4-5		22	C4-21	
7	C4-6		23	C4-22	
8	C4-7		24	C4-23	
9	C4-8		25	C4-24	
10	C4-9		26	C4-25	
11	C4-10		27	C4-26	
12	C4-11		28	C4-27	
13	C4-12		29	C4-28	
14	C4-13				
15	C4-14				
16	C4-15				

Table 2MIC, ClogP, and ATP IC₅₀ values of the compounds **C4**, **C4-1** to **C4-28** against *Mtb* mc²6230.


Compound code	<i>Mtb</i> mc ² 6230 MIC (μg/mL)	<i>Mtb</i> mc ² 6230 MIC (μM)	ClogP	<i>Mtb</i> mc ² 6230 ATP IC ₅₀ (μg/mL)	<i>Mtb</i> mc ² 6230 ATP IC ₅₀ (μM)
C4	4	11.476	5.24	9.35 ± 1.11	26.82 ± 1.11
C4-1	4	13.042	3.80	6.95 ± 0.21	22.70 ± 0.21
C4-2	16	49.92	4.12	32.60 ± 0.12	101.72 ± 0.12
C4-3	32	99.84	4.68	11.82 ± 0.01	36.87 ± 0.01
C4-4	64	176.52	5.80	50.29 ± 0.23	156.90 ± 0.23
C4-5	64	159.76	5.87	13.35 ± 0.36	33.32 ± 0.36
C4-6	64	207.48	4.27	12.63 ± 0.05	40.94 ± 0.05
C4-7	32	99.22	4.80	30.10 ± 1.13	93.42 ± 1.13
C4-8	16	47.54	5.33	29.77 ± 0.09	88.46 ± 0.09
C4-9	16	43.88	6.39	18.41 ± 0.09	50.49 ± 0.09
C4-10	32	99.22	4.67	17.87 ± 0.10	55.41 ± 0.10
C4-11	8	24.80	4.70	8.71 ± 1.11	27.00 ± 1.11
C4-12	64	178.51	5.19	52.29 ± 1.23	145.84 ± 1.23
C4-13	32	85.90	5.27	27.89 ± 0.08	74.87 ± 0.08
C4-14	64	177.54	5.59	36.14 ± 1.18	100.25 ± 1.18
C4-15	128	311.83	6.53	31.24 ± 1.09	76.10 ± 1.09
C4-16	128	300.13	6.38	22.16 ± 0.02	51.96 ± 0.02
C4-17	>128	>326.08	6.37	31.28 ± 1.11	79.68 ± 1.11
C4-18	128	343.57	5.27	18.25 ± 1.14	48.98 ± 1.14
C4-19	>128	>366.24	4.43	17.39 ± 1.05	49.75 ± 1.05
C4-20	128	356.04	4.52	18.54 ± 1.60	51.57 ± 1.60
C4-21	128	342.71	4.96	55.35 ± 0.04	148.19 ± 0.04
C4-22	64	177.06	4.78	11.52 ± 0.09	31.87 ± 0.09
C4-23	128	342.67	3.60	27.27 ± 1.06	73.00 ± 1.06
C4-24	64	162.63	5.70	27.40 ± 1.31	69.62 ± 1.31
C4-25	128	325.26	5.70	16.66 ± 0.08	42.33 ± 0.08
C4-26	64	191.33	4.56	16.13 ± 0.72	48.22 ± 0.72
C4-27	64	190.19	3.76	22.28 ± 1.57	66.21 ± 1.57
C4-28	32	90.77	4.50	22.30 ± 0.54	63.25 ± 0.54
CBR-5992	0.5	1.57	2.06	0.14 ± 0.08	0.44 ± 0.08
Isoniazid	0.06	0.43	−0.66		
Rifampicin	0.03	0.036	3.71		

Table 2 (continued)


Compound code	<i>Mtb</i> mc ² 6230 MIC (μg/mL)	<i>Mtb</i> mc ² 6230 MIC (μM)	ClogP	<i>Mtb</i> mc ² 6230 ATP IC ₅₀ (μg/mL)	<i>Mtb</i> mc ² 6230 ATP IC ₅₀ (μM)
Clofazimine (CLFZ)	0.25	0.52	7.54		
BDQ	n.d	n.d	n.d	0.051 ± 0.02	0.09 ± 0.03
Rotenone (ROT)	>64	>162.26	4.20		

Table 3Determination of ATP IC₅₀ (μg/mL, and μM) against NS *Mtb* mc² 6230.

Serial no.	Compound code	NS <i>Mtb</i> mc ² 6230 ATP IC ₅₀ (μg/mL)	NS <i>Mtb</i> mc ² 6230 ATP IC ₅₀ (μM)
1	C4-1	5.99 ± 1.45	19.56 ± 1.45
2	C4-11	4.91 ± 1.83	15.22 ± 1.83
3	CBR-5992	0.30 ± 0.08	0.94 ± 0.08
4	BDQ	0.02 ± 0.007	0.04 ± 0.007

2.6. Time kill kinetic experiment

To conceive the dynamics of the lethal response of the test active compounds (**C4-1**, and **C4-11**), with respect to dose over time, kill kinetic experiment was performed. Compounds were evaluated against aerobically cultured, replicating strain of H37Rv mc²6230 strain. Bacterium were exposed to 1 ×, 2 ×, and 4 × of their MIC, over a time period of 15 days; and response were recorded on 5th, 10th, and 15th day of the incubation, post-compound treatment. Both the compounds have exhibited bactericidal response at 4 × MIC, on 10th, and 15th day of incubation, with 1.5 log₁₀ reduction in growth (Fig. 5).

2.7. Peredox-mCherry experiment to evaluate the NDH-2 selectivity

NDH-2 is the primary dehydrogenase enzyme that *Mtb* uses to channel electrons in the ETC. It functions by oxidizing NADH to NAD⁺ while simultaneously reducing menaquinone to menaquinol. Inhibiting NDH-2 results in an immediate increase in intracellular NADH level. To assess the specificity of our test compounds (**C4-1**, and **C4-11**) for NDH-2, we employed the Peredox-mCherry expressing *Mtb* strain. The binding of the inhibitors to NDH-2 is reflected by an elevated level of free NADH, which subsequently binds to the circularly permuted T-sapphire sensor, causing an increase in green fluorescence. The enhancement of green fluorescence in the groups treated with NDH-2 inhibitors compared to the control group serves as an indicator of the target specificity of these inhibitors for NDH-2. Peredox-mCherry expressing *Mtb* strain, on exposure to compounds **C4-1** and **C4-11**, at concentrations of 1 ×, 2 ×, and 4 × of their MIC for 24 h, have experienced a significantly altered NADH: NAD⁺ ratio, similar to that of the reported NDH-2 inhibitor **CBR-5992** (Fig. 6). On treatment with NDH-1 inhibitor rotenone (ROT), a partially altered NADH: NAD⁺ ratio was observed.

2.8. Susceptibility, cross-resistance, and single nucleotide polymorphism in NDH-2 Gene

To ensure a deeper understanding of the site of action of our test active compounds **C4-1** and **C4-11**, they were assayed for susceptibility against the cross-resistant mutants to the reported NDH-2 inhibitor **CBR-**

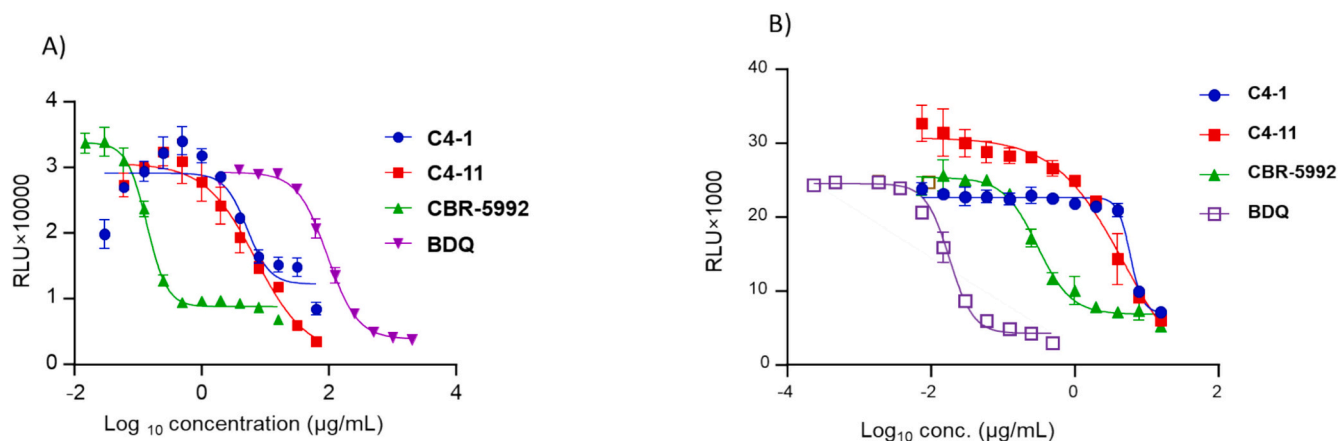


Fig. 2. Determination of ATP IC_{50} ($\mu\text{g/mL}$) against: A) replicating *Mtb mc*² 6230, B) against non-replicating, nutrient starved *Mtb mc*² 6230. When the derivative strains of *Mtb mc*² 6230 were exposed to ten two-fold serial dilutions for 24 h, significant ATP depletion was observed.

Table 4

MIC of C4, C4-1, and C4-11 against the drug resistant *Mtb* strains.

Comp. Code	MIC					
	mc ² 8243	mc ² 8243	mc ² 8247	mc ² 8247	mc ² 8259	mc ² 8259
	($\mu\text{g/mL}$)	(μM)	($\mu\text{g/mL}$)	(μM)	($\mu\text{g/mL}$)	(μM)
C4	4	11.47	4	11.47	4	11.47
C4-1	8	26.13	4	13.06	4	13.06
C4-11	8	24.80	8	24.80	4	12.40
CBR-5992	0.125	0.39	0.250	0.78	0.125	0.39
Isoniazid	>128	>824.90	0.25	1.61	>128	>824.90
Rifampicin	0.12	0.14	>128	>155.53	>128	>155.53

5992. Strains with any impaired NDH-2 gene composition should remain viable even after exposure to our test active NDH-2 inhibitors. For the identification of NDH-2 resistant *Mtb* strains; we initially selected six colonies and grew them to the log phase. Among these six colonies, only two mutants, C^{r-3} and C^{r-5}, had exhibited more than eightfold resistance to literature-reported NDH-2 inhibitor **CBR-5992**. To find the mutations in the gene that codes for NDH-2, these two mutants were also sequenced. The results of the sequencing process showed that NDH-2 had amino acid alterations at various locations due to point mutations at multiple locations in the gene [37,43]. These two resistant mutants showed increase in MIC when treated with compounds **C4-1**, and **C4-11**; thereby validating the target specificity of both the

derivatives to NDH-2 (Table 5).

2.9. Evaluation of cell cytotoxicity

The test active compounds **C4-1**, and **C4-11** were subjected to MTT assay; for evaluating their cytotoxic response and for learning about their therapeutic window. The HepG2 cell line was utilized for the toxicity study, and the strongly cytotoxic anti-cancer drug Tamoxifen was used as a positive control in the assay. The cytotoxicity concentration (CC_{50}) and safety index (SI) are mentioned in Table 6, where the assays have been repeated in triplicate. The SI of all the derivatives were in the acceptable range, as all of them displayed $SI > 10$ [37, 42a].

3. Computational analysis

3.1. Molecular docking study

To acquire an insight into the mode of interaction of the most active compounds of the series (**C4-1**, and **C4-11**), at the receptor-ligand interface; *in silico* docking study was conducted post-synthesis and *in vitro* analysis. For that, we have availed the protein structure of the mycobacterium NDH-2, with PDB ID: 9MQZ [51]. Docking was performed in the prior-fixed search place of 2-mercapto-quinazolinone inhibitor. The docking was carried out using AutoDock Vina version 1.1.2 [43]. The binding interactions are illustrated in Fig. 7, and the interacting amino acid residues are listed in Table 7.

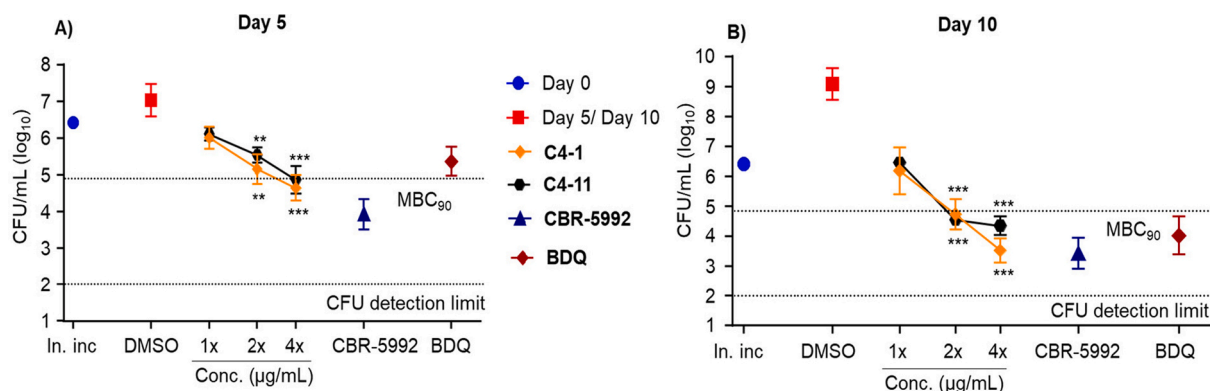


Fig. 3. MBC determination of compounds **C4-1**, and **C4-11**: the killing efficacy response of the compounds **C4-1**, and **C4-11** against the replicating strain of *Mtb H37Rv mc*² 6230. 90 % bacterial killing, i.e., MBC_{90} is indicated by the first black dotted line w.r.t to the initial inoculum. Significant difference was determined by using one-way ANOVA followed by Dunnett's test. All the values are expressed as mean $\pm$ SDs ($n = 3$), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs initial inoculum (day 0). **CBR-5992**: 1.25 $\mu\text{g/mL}$ and **BDQ**: 0.5 $\mu\text{g/mL}$ have been used as a positive control in this assay.

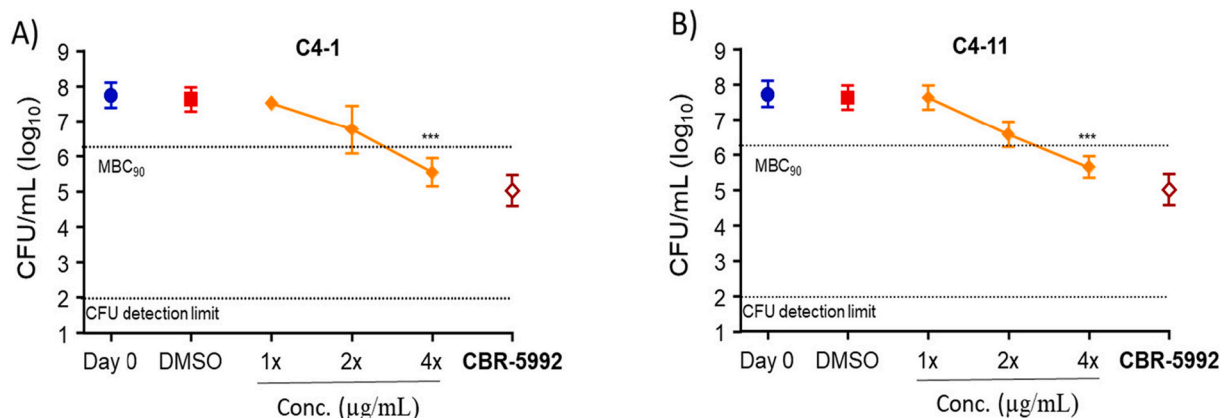


Fig. 4. Determination of MBC against nonreplicating nutrient starved mc^26230 . Significant difference was determined by using one-way ANOVA followed by Dunnett's test. All the values are expressed as mean $\pm$ SDs ($n = 3$), *** $p < 0.001$ vs initial inoculum (day 0), **CBR-5992**: 1.25 $\mu\text{g/mL}$ was used as a positive control in this assay.

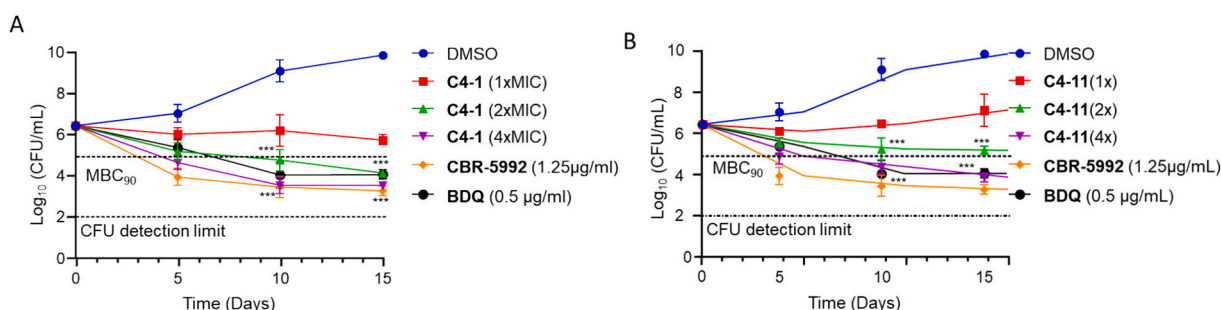


Fig. 5. Kill kinetic graphs of: A) **C4-1**, and B) **C4-11**. **CBR-5992** (1.25 $\mu\text{g/mL}$), the literature reported NDH-2 inhibitor was used as a positive control, along with **BDQ** (0.5 $\mu\text{g/mL}$). The data are presented as mean $\pm$ SDs ($n = 3$) in triplicates for each concentration of compounds. The significance difference was determined using two-way ANOVA followed by Tukey's test. *** $p < 0.001$ vs. Day 0 initial inoculum.

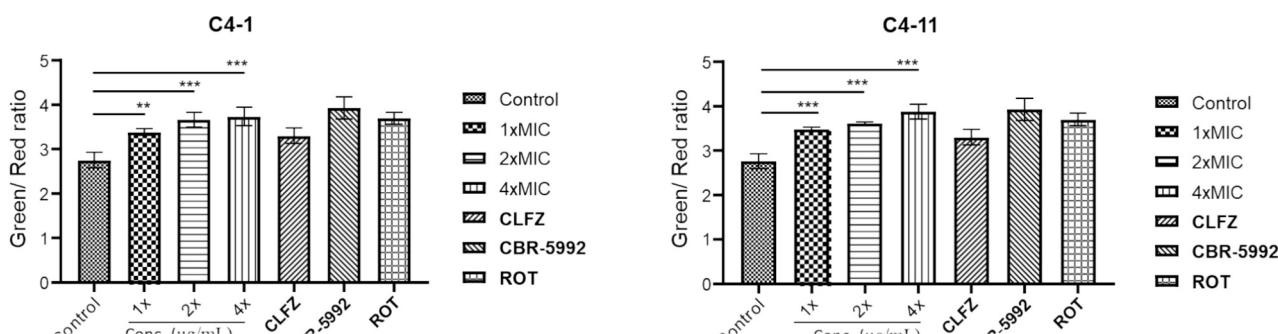


Fig. 6. Peroxide-mCherry experiment: a hike in the green to red fluorescence ratio signifies an increase in $\text{NADH} : \text{NAD}^+$ ratio, on exposure to **C4-1**, and **C4-11** for 24 h. Significant difference was determined by using one-way ANOVA followed by Dunnett's test. All the values are expressed as mean $\pm$ SDs ($n = 3$), ** $p < 0.01$, *** $p < 0.001$ vs control, **CBR-5992**: 0.625 $\mu\text{g/mL}$; **CLFZ**: 2.5 $\mu\text{g/mL}$; and **ROT**: 60 $\mu\text{g/mL}$ have been used as a positive control in this assay. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

From the binding interactions, as depicted in the Fig. 7; comprehension of the probable type of interactions between the amino acid residues residing at the binding site and the various motifs of the synthesized derivatives, necessity for effective binding was obtained. **C4-1**, and **C4-11**, have displayed similar kind of binding interactions, engaging the same amino acid residues at the binding site. We see, that the delocalized pi-electron cloud of the benzothiazole core in **C4-1**, and **C4-11**, have participated in multiple interactions, by utilizing its pi-electron cloud. Pi-alkyl interactions were noted with the amino acid residues Met^{A:245}, Ala^{A:275}. Pi-pi stacking was observed between the phenyl ring electron cloud of PHE^{A:123}, and benzothiazole core. Ala^{A:275} generated pi-sigma interaction; and Ala^{A:213} has formed pi-donor H-

bond with the benzothiazole electron cloud as well. Conventional H-bond interaction was noted between the carbonyl oxygen, and the following two amino acid residues, namely, Ala^{A:175}, Gly^{A:176}. Additionally, the nitrogen atom of the thiazole ring displayed pi-donor H bond interaction with Asp^{A:212}.

C4 displayed pi-sigma interactions with Pro^{A:177}, Val^{A:329}; pi-pi stacking with the co-factor FAD, via the delocalized pi-electron cloud of the benzothiazole core. The cyclohexyl moiety of the acetamide side chain has participated in pi-alkyl interactions with FAD and Lys^{A:136}. FAD has also displayed conventional H-bond interaction with the carbonyl group of the side motif as well.

From the observations of the *in-silico* findings we may state that: the

Table 5

Determination of susceptibilities of mutants, cross-resistance to test drugs and mutations in NDH-2 encoding gene.

Mutant code	MIC values of test molecules against the mutants raised in µg/ml (fold change in MIC w.r.t. susceptible strain)			Change in amino acids in NDH2 based on SNP confirmation
	CBR-5992	C4-1	C4-11	
C ^{r-3}	2 (16)	>128 (>16)	>128 (>8)	His-400-Cys; Ser-283-Gly; Trp-396-Arg; ala-333-Val; Leu-399-Gly
C ^{r-5}	1 (8)	>128 (>16)	>128 (>8)	Ser-283-Gly; Trp-396-Arg; Leu-285-Trp;

Table 6

Cytotoxicity and selective index profile of the test NDH-2 inhibitors C4-1, and C4-11.

Molecule code	CC ₅₀ (µg/mL)	CC ₅₀ (µM)	SI (CC ₅₀ /MIC)
C4	>50	>143.46	26.48
C4-1	>50	>163.35	16.52
C4-11	>50	>318.68	12.85
CBR-5992	>50	>157.52	>200
Tamoxifen	4.766	12.82	

core benzothiazole drug scaffold is majorly engaged in binding interactions, and remain embedded in the hydrophobic pocket; while the acetamide side chain is projected outwards. The thioether linkage does not produces any considerable interactions. Reducing the alicyclic ring size, and further ring opening of the acetamide side chain led to elimination of any possible alkyl interactions of the side chain.

3.2. Molecular dynamics (MD) simulation analysis

To evaluate the stability of the protein-ligand complex formation, and conceive the mechanism of interaction at the binding site; MD simulation was performed of the following compounds, namely, **CBR-5992**, **C4**, **C4-1**, and **C4-11**. While analysing various parameters, the first docking pose of the protein-ligand complex was employed as the starting point, and the MD analysis was performed for 100 nano seconds (ns), using GROMACS software.

The RMSD (Root Mean Square Deviation) of all the analysed compounds and NDH-2 gave an insight of the structural deviation of the protein ligand complex from a reference structure over time, during simulation. The RMSD plot of the simulated systems (Fig. 8A) display a rapid rise in RMSD value, in the initial 10 ns, arising from structural rearrangement. The initial hike in RMSD is followed by, gradual equilibration over time. All the four systems exhibited RMSD values within 0.5 nm; indicative of a stable system; and to steady rise in RMSD after initial 10 ns further confirmed that overall fold of the protein remained stable throughout. The RMSD graph of all the ligands is mentioned in Fig. 8B. The RMSD fluctuation in the 5–10 ns, of **CBR-5992** is indicative of conformational changes and shift in binding mode in the initial period of MD run. The most stable protein-ligand interaction was attained by **C4-1**; with RMSD value ranging between 0.1 and 0.25 nm, accompanied by minimum fluctuation throughout. While **C4** displayed a stable binding, with moderate fluctuation in the entire 100 ns; **C4-11** displayed the most conformational flexibility.

The root mean-square fluctuations (RMSF) of the simulated systems (Fig. 8C), showcases that the core of the protein is relatively rigid, indicated by relatively low RMS value between 50 and 400 residues (approximately). The central rigid portion is accompanied by loops impregnated flexible regions in the N-terminal, and C-terminal of the protein. The interaction with **CBR-5992** appears to enhance the stability of the protein, whereas the alternative scenarios (with **C4**, **C4-1**, **C4-11**)

facilitate increased flexibility, especially within the terminal and loop regions. **C4-1**, **C4-11** exhibit relatively higher RMSF values in the flexible N-terminal, C-terminal regions, indicating that the protein is more dynamic with those ligands, compared to when bound to **CBR-5992**. **C4** also displayed similar flexibility profile like **C4-1**, and **C4-11**. The RMSF values of the ligands are presented in Fig. 8D. An overall low RMSF value of all the four ligands, confirms the overall fair stability of them within the protein macromolecule, at the binding site. While the RMSF values of **CBR-5992**, **C4**, and **C4-1** ranged between 0.025 and 0.2 nm; a slightly higher RMSF of 0.27 nm was displayed by the terminal portion of the **C4-11**.

The radius of gyration (Rg) is a visual presentation of the simulation trajectory. It provides measurement of the spatial extent of the system under investigation. A decrease in Rg arises from protein folding, and an increase in Rg is due to protein unfolding. Rg of all the four complexes are presented in Fig. 9A. The graph demonstrates that the **CBR-5992** exhibits a relatively stable overall size and morphology throughout the simulation. The variations in the radius of gyration (Rg) for configurations **C4**, **C4-1**, and **C4-11** indicate that the molecule is not perfectly spherical but possesses an anisotropic shape, exhibiting dependency on direction. The consistency of these Rg values over the course of the simulation implies that the molecule does not experience significant conformational changes that would markedly affect its overall dimensions or extension along its principal axes within the observed timeframe.

The solvent accessible surface area (SASA) is a visual representation of the surface area of a macromolecule, that is accessible to a water. High SASA values indicate hydrophobic region, complementing protein unfolding; thereby increasing the available surface area for solvent access. The SASA analysis highlights the dynamic characteristics of molecular systems under investigation, within their solvent milieu. The variations in average SASA values indicate differences in the extent of solvent exposure among the systems studied. Notably, **C4-11** exhibits the highest degree of solvent exposure, whereas **CBR-5992**, **C4**, and **C4-1** show comparable, albeit lower, average solvent accessibility (Fig. 9B). The observed fluctuations in SASA are indicative of ongoing conformational changes and potential interactions with the solvent across all systems during the simulation period.

Determining the number of H-bond formation in protein-ligand complex formation, during simulation gives an insight of the stability of the complex. A constant number of H-bond throughout the analysis time is an indication of well folded protein structure. The highest number of H-bond interactions were displayed by **CBR-5992**, which ranges between 2 and 4. The number has further increased to 5, in the initial 20–40 ns, suggests that **CBR-5992** consistently forms a significant number of stable interactions via hydrogen bonds. For **C4-1**, the number of H-bond formation is intermediate between **CBR-5992** and **C4**, **C4-11**; displaying a continuous 1H-bond count, followed by occasional hike to 3, and dips to 0H-bond. **C4**, produced fluctuating number of H-bond between 0 and 2 throughout. The lowest number of H-bond formation was noted for **C4-11** (Fig. 9C).

3.3. In silico physicochemical properties and ADME profiling

The *in silico* physicochemical properties and pharmacokinetic profile of our bioactive compounds **C4**, **C4-1**, and **C4-11** were evaluated using SwissADME, and pkCSM web servers [44,45]; data of all the analysed parameters are mentioned in Table 8. We may state that the compounds would display a good absorption profile in the intestinal tract, as all of them have predicted % absorption value more than 30 %. None of the compounds serves as the substrate of the ATP binding cassette transporter glycoprotein P-gp glycoprotein. While the estimated volume of distribution of **C4-1** was low, **C4** and **C4-11** possess a good distribution profile (as log VDss > -0.15 for **C4-11**). All of our compounds displayed poor blood-brain-barrier (BBB) and weak CNS permeability; as is indicated by the predicted numerical values of log BB value <0.3 and logP

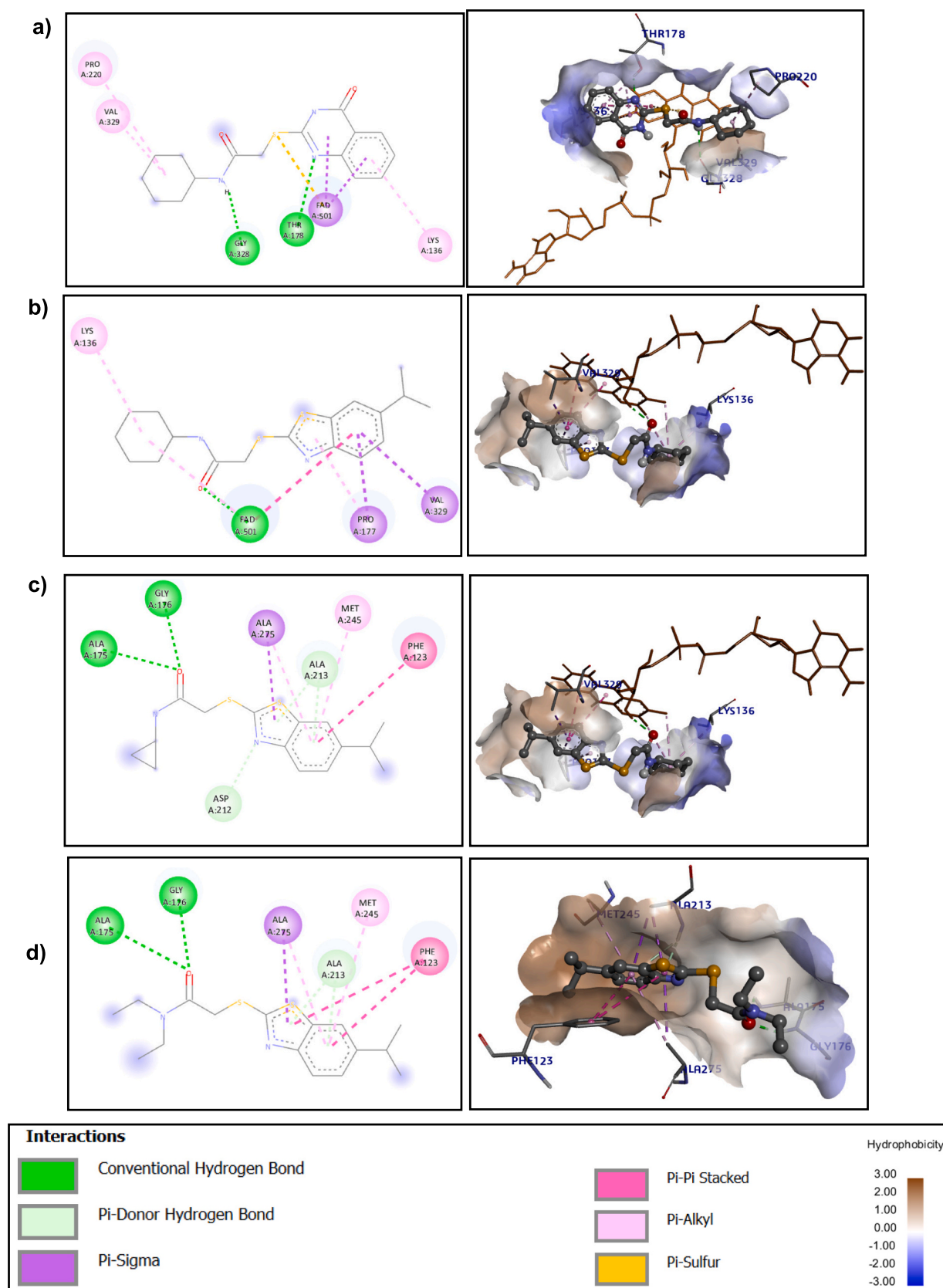


Fig. 7. Binding interactions of compounds (a) CBR-5992, (b) C4, (c) C4-1, and (d) C4-11 in the hydrophobic surface of receptor.

Table 7

Binding affinity values of docked NDH-2 inhibitors: **CBR-5992**, **C4**, **C4-1**, and **C4-11**.

Compound code	Binding affinity (kcal/Mol)	Interacting residues
CBR-5992	−8.1	Lys ^{a:136} , Thr ^{a:178} , Pro ^{a:220} , Gly ^{a:328} , Val ^{a:329} , FAD ^{a:501}
C4	−8.0	Lys ^{a:136} , Pro ^{a:177} , Val ^{a:329} , FAD ^{a:501}
C4-1	−8.2	Phe ^{a:123} , Ala ^{a:175} , Gly ^{a:176} , Asp ^{a:212} , Ala ^{a:213} , Met ^{a:245} , Ala ^{a:275}
C4-11	−7.0	Phe ^{a:123} , Ala ^{a:175} , Gly ^{a:176} , Ala ^{a:213} , Met ^{a:245} , Ala ^{a:275}

value < −2. All of our compounds serve as substrate of the phase I drug metabolism enzyme CYP3 A4; therefore, care should be taken while dose assessing. None of them act as either substrate of CYP2D6, and inhibitor of CYP3 A4. No derivatives do not display any clearance liability from their excretion profile, which include combination of renal and hepatic clearance.

The oral bioavailability radar plot of our compounds **C4-1**, and **C4-11** indicate that both of our synthesized derivatives share a favourable predicted oral bioavailability parameter; as both the derivatives fall within the hexagonal pink zone (Fig. 10). The pink zone delineates the optimal range for each descriptor's flexibility, which are as follows: should not exceed 9 bonds; saturation, which must not be

lower than 0.25; solubility, limited to a maximum of 6; topological polar surface area (TPSA), constrained to a range of 20 Å² to 130 Å²; molecular size, defined by a molecular weight between 150 and 500 g mol^{−1}; and lipophilicity, indicated by XLOGP3 values ranging from −0.7 to +5.0. The parent compound **C4**, on the other hand displays deviation from favourable behaviour in lipophilicity parameter; with XlogP3 > 3.5. Compounds residing with the hexagonal pink area are assumed to possess optimum lipophilicity, solubility, size, saturation, topological polar surface area (TPSA) properties [52]. Our compounds have the TPSA value within the acceptable range of 20–130; and produces no violation of Lipinski's rule of five [53]. Additionally, both the compounds display zero PAIN (pan-assay interference compounds: it refers to the compounds possessing sub-structures, that may display false positive results, following interaction with multiple irregular biological targets during high throughput screening) alert [54].

4. Conclusion

Identification of novel anti-TB compounds with potency to treat DR-TB is strictly essential to reduce the global TB burden. TB not only leads to economic loss but also causes loss of life when treatment is inadequate. Since TB is a communicable disease, spread through sneezing and coughing from an infected individual to a healthy person, prompt diagnosis and treatment are essential. Building on our previous work in developing potent *Mtb* NDH-2 inhibitors; we have explored the effect of

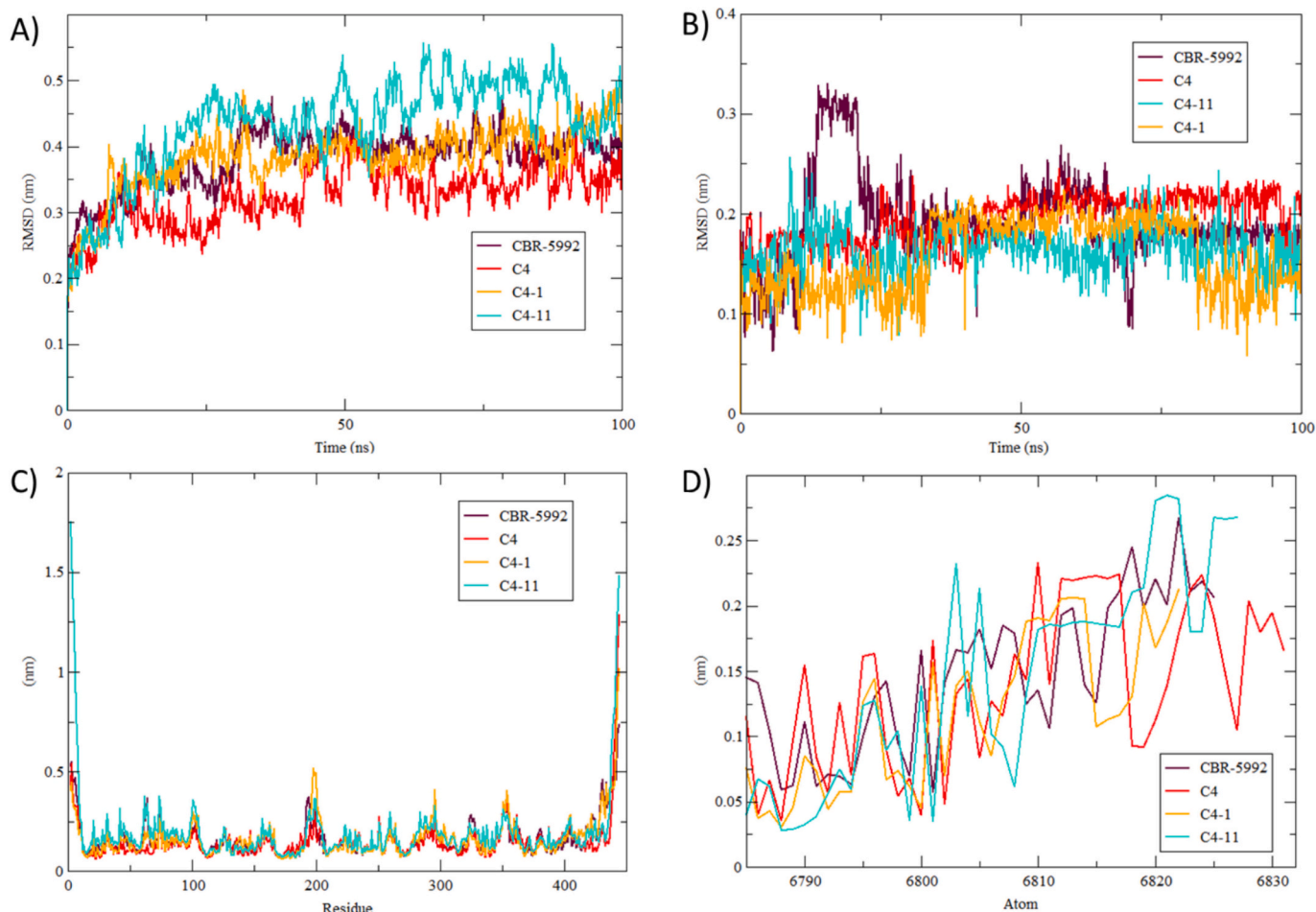


Fig. 8. A) RMSD of the protein NDH-2 (PDB ID: 9MQZ) during the complex formation with **CBR-5992** (maroon), **C4** (red), **C4-1** (orange), and **C4-11** (turquoise); b) RMSD of the ligands: **CBR-5992** (maroon), **C4** (red), **C4-1** (orange), and **C4-11** (turquoise) during the complex formation with protein NDH-2 (PDB ID: 9MQZ); a) RMSF of the protein NDH-2 (PDB ID: 9MQZ) during the complex formation with **CBR-5992** (maroon), **C4** (red), **C4-1** (orange), and **C4-11** (turquoise); b) RMSF of the ligands: **CBR-5992** (maroon), **C4** (red), **C4-1** (orange), and **C4-11** (turquoise) during the complex formation with protein NDH-2 (PDB ID: 9MQZ). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

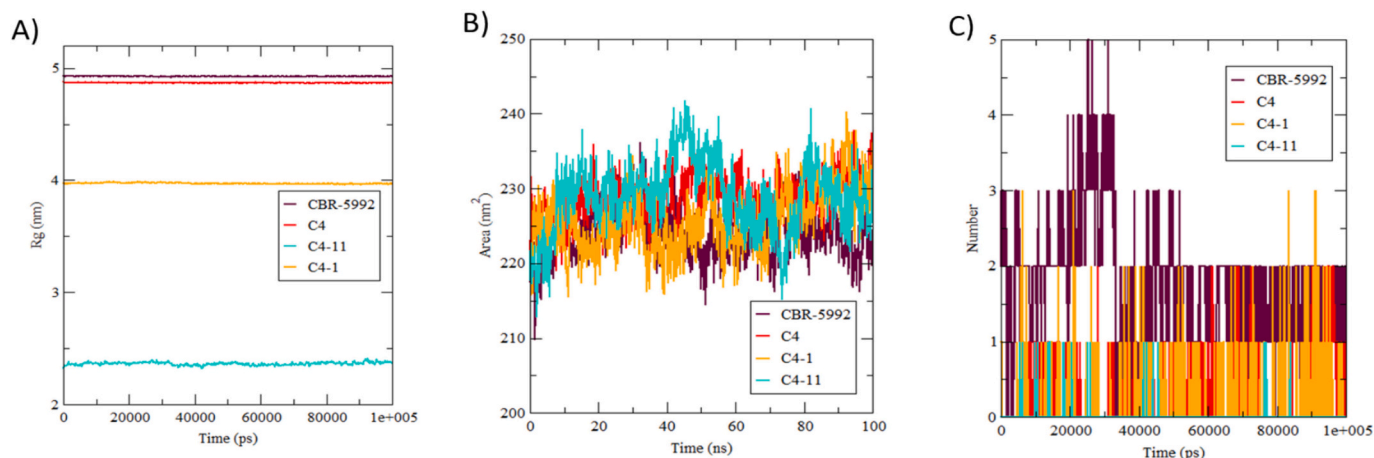


Fig. 9. A) Radius of gyration (RoG) for **CBR-5992** (maroon), **C4** (red), **C4-1** (orange), and **C4-11** (turquoise) during the complex formation with protein NDH-2 (PDB ID: 9MQZ); B) Changes in the total SASA of **CBR-5992** (maroon), **C4** (red), **C4-1** (orange), and **C4-11** (turquoise) during the complex formation with protein NDH-2 (PDB ID: 9MQZ); C) Average hydrogen-bond number (number) formed between NDH-2 and **CBR-5992** (maroon), **C4** (red), **C4-1** (orange), **C4-11** (turquoise) during the complex formation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 8
ADME and Drug-likeness Properties of compounds **C4**, **C4-1**, and **C4-11**.

Compound code		C4	C4-1	C4-11
Absorption	Human intestinal absorption numeric (%)	91.439	92.162	91.408
	absorbed)			
Distribution (human)	P-gp substrate	No	No	No
	Volume of distribution	0.543	0.383	0.382
	BBB permeability numeric (log BB)	0.042	-0.012	0.083
	CNS permeability numeric (log PS)	-2.513	-2.179	-2.149
Metabolism	CYP3A4 substrate	Yes	Yes	Yes
	CYP2D6 substrate	No	No	No
	CYP3A4 inhibitor	No	No	No
Excretion	Total clearance numeric (log mL/min/kg)	0.246	0.214	0.459
Physicochemical properties	No. of H-bond donor	1	1	0
	No. of H-bond acceptors	2	2	2
	TPSA (Å²)	95.53	95.53	86.74
	Lipinski	Yes	Yes	Yes
Medicinal chemistry	PAINS	0 alert	0 alert	0 alert

introducing a variety of substitutions to the side chain acetamide moiety of 2-mercaptobenzothiazole derivatives, while maintaining a favourable isopropyl substitution at the C6 position of the benzothiazole core. Through comprehensive *in vitro* assays and computational analysis of the synthesized derivatives, we observed that **C4-1** derivative, which features a cyclopropyl ring substituted acetamide side chain displayed marginally high MIC (in μM), with improved ATP inhibitory response than the parent compound **C4**. This identified NDH-2 inhibitors produce bactericidal response, with NDH-2 target specificity, and equal potency against mono and multi-drug resistant *Mtb* strain. SI falls within the permissible range for both the derivatives. We infer that the *in vitro* findings together with the computational results, will aid in the future design of NDH-2 inhibitors by providing valuable insights into the pharmacophoric features that need to be considered in order to enhance inhibitory efficacy, in the absence of a reported crystal structure of *Mtb* NDH-2.

5. Experimental

5.1. Materials and methods

All chemicals were purchased from Sigma-Aldrich (St. Louis, USA),

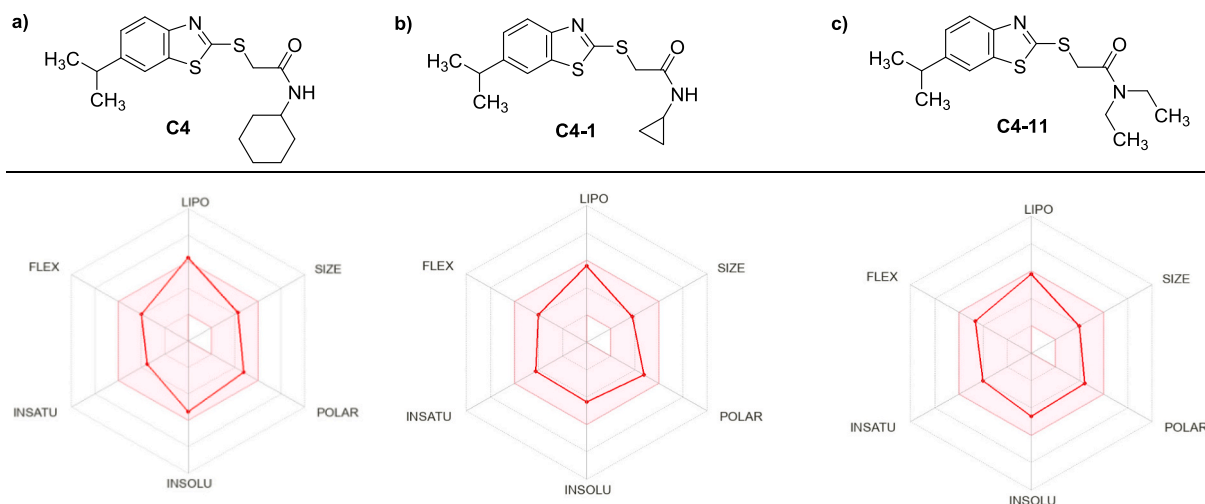


Fig. 10. Bioavailability radar of compounds a) **C4**, b) **C4-1**, and c) **C4-11**; the hexagonal pink zone indicates the area of optimal physicochemical properties for oral bioavailability. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

TCI Chemicals (Tokyo, Japan), SRL Chemicals (Mumbai, India), and Avra (Hyderabad, India), and used as received. Molychem silica gel (Mumbai, India) (60–120 mesh) was used for column chromatography, and thin-layer chromatography was performed on Merck pre-coated silica gel 60-F254 plates. All other chemicals and solvents were obtained from commercial sources and purified using standard methods. The ^1H NMR and ^{13}C NMR spectra were recorded on Bruker Advance spectrometers. Data are represented as follows: chemical shift, integration, multiplicity (br = broad, s = singlet, d = doublet, dd = double doublet, t = triplet, q = quartet, dd = doublet of doublet, dt = doublet of triplet, td = triplet of doublet and m = multiplet), and coupling constants in hertz (Hz).

6. Chemistry

Procedure for the synthesis of 6-isopropyl, benzo[d]thiazole-2-thiol (A):

An oven-dried round bottom flask was charged with 4-isopropyl, 2-bromo aniline (5 mmol), potassium ethyl xanthate (1.5 equiv.) in anhydrous DMF, under nitrogen atmosphere. The reaction mixture was stirred at 165 °C for 6 h, under the mentioned inert atmosphere. On completion, the reaction mixture was cooled to room temperature and diluted with water (8 mL), and acidified to pH 3–4, by adding 1 N HCl. The formed suspension was subjected to uniform stirring for 30 min, followed by filtration. The crude product (A) was collected by filtration under vacuum.

Procedure for the synthesis of 2-bromo-acetamide derivatives (B):

An oven-dried round bottom flask was charged with EDC.HCl (1 mmol), bromo acetic acid (1 equiv.), and the corresponding amine (1 equiv.), in DCM. The reaction mixture was stirred at room temperature for 30 min. After completion of reaction, the reaction mixture was extracted with ethyl acetate (15 mL $\times$ 3). The combined organic layers were washed with brine, dried over Na_2SO_4 , and concentrated in vacuum to acquire the intermediate B.

Procedure for the synthesis of 6-isopropyl, 2-(benzo[d]thiazol-2-ylthio)acetamide derivatives (C4, C4-1 to C4-28):

An oven-dried round bottom flask was charged with the 6-isopropyl, benzo[d]thiazole-2-thiol (A) (1 mmol) and the respective 2-bromoacetamide derivatives (B) (1 equiv.) in DMF. The reaction mixture was stirred at room temperature for 2–4 h. Next, DIPEA (0.5 equivalent) was added in the ongoing reaction, and the reaction mixture was allowed to be stirred overnight. After completion of reaction, the reaction mixture was extracted with ethyl acetate (15 mL $\times$ 3). The combined organic layers were washed with brine, dried over Na_2SO_4 , and concentrated in vacuum. The residue was purified by column chromatography on silica gel to afford the desired products (C4, C4-1 to C4-28).

N-cyclohexyl-2-((4-oxo-3,4-dihydroquinazolin-2-yl)thio)acetamide (CBR-5992):

^1H NMR (DMSO- d_6 , 500 MHz) δ = 12.68 (s, 1H, -NH), 8.17 (d, J = 8 Hz, Ar-H), 7.79–7.76 (m, 1H, Ar-H), 7.49 (d, J = 8.5 Hz, 1H, Ar-H), 7.44–7.41 (m, 1H, Ar-H), 3.93 (s, 2H, -CH₂), 3.56–3.50 (m, 1H, Cyc CH), 1.74–1.71 (m, 2H, Cyc -CH₂), 1.68–1.65 (m, 2H, Cyc -CH₂), 1.55–1.52 (m, 1H, Cyc -CH₂), 1.26–1.14 (m, 5H, Cyc 3-CH₂); ^{13}C NMR (DMSO- d_6 , 125 MHz) δ = 166.4 (CO), 161.6 (Ar C), 155.9 (Ar C), 148.8 (Ar C), 135.1 (Ar C), 126.5 (Ar C), 126.3 (Ar C), 126.2 (Ar C), 120.4 (Ar C), 48.4 (-Cyc C), 37.7 (-Cyc 2C), 34.6 (-CH₂), 25.6 (-Cyc C), 24.9 (Cyc-2C).

N-cyclohexyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4):

Melting point: 108–112 °C. IR (KBr) cm^{-1} : 3291, 2929, 2854, 2362, 2339, 1723, 1675, 1633, 1536, 1444, 1403, 1333, 1275, 1257, 1241, 1151, 1125, 1084, 1055, 1008, 979, 962, 929, 892, 876, 856, 830, 804, 782, 718, 705, 661, 620. ^1H NMR (DMSO- d_6 , 500 MHz) δ = 8.21 (d, J = 7.5 Hz, 1H, -NH), 7.84 (d, J = 1.5 Hz, 1H, Ar-H), 7.23 (d, J = 8.5 Hz, 1H, Ar-H), 7.36 (d, J = 1.5, 4.25 Hz, 1H, Ar-H), 4.09 (s, 2H, -CH₂), 3.56–3.53 (m, 1H, -CH), 3.03–2.98 (m, 1H, -CH), 1.75–1.72 (m, 2H, -CH₂), 1.67–1.65 (m, 1H, -CH₂), 1.55–1.52 (m, 1H, -CH₃), 1.30–1.12 (m, 12H, -CH₃); ^{13}C NMR (DMSO- d_6 , 125 MHz) δ = 165.65 (CO), 165.57 (Ar

C), 151.43 (Ar C), 145.62 (Ar C), 135.39 (Ar C), 125.73 (Ar C), 121.15 (Ar C), 119.36 (Ar C), 48.45 (-Cyc C), 37.32 (-CH₂), 33.90 (-CH), 32.66 (-Cyc 2C), 25.61 (-Cyc C), 24.82 (Cyc-2C), 24.49 (-CH₃ 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{18}\text{H}_{25}\text{N}_2\text{OS}_2$: 349.1408; found: 349.1402.

N-cyclopropyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-1):

Melting point: 105–108 °C. IR (KBr) cm^{-1} : 3981, 3960, 3894, 3879, 3451, 3388, 3265, 3132, 3104, 3053, 3029, 3011, 2960, 2924, 2904, 2870, 2854, 2715, 2674, 2357, 2332, 2313, 1725, 1549, 1479, 1449, 1405, 1380, 1313, 1257, 1233, 994, 922, 892, 786, 724, 667, 626. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.75 (d, J = 8.4 Hz, 1H, Ar-H), 7.69 (s, 1H, NH), 7.61 (d, J = 1.8 Hz, 1H, Ar-H), 7.32 (dd, J = 8.4 Hz, 1.8 Hz, 1H, Ar-H), 3.89 (s, 2H, -CH₂), 3.06–2.99 (m, 1H, -CH), 2.75–2.71 (m, 1H, Cyc-H), 1.30 (d, J = 7.2 Hz, 6H, 2CH₃), 0.76–0.72 (m, 2H, Cyc-H), 0.47–0.45 (m, 2H, Cyc-H); (CDCl_3 , 150 MHz) δ = 169.77 (CO), 165.34 (Ar C), 150.68 (-Ar C), 146.20 (Ar C), 135.67 (Ar C), 125.50 (Ar C), 120.92 (Ar C), 118.53 (Ar C), 36.04 (-CH₂), 34.21 (-CH), 24.17 (Cyc-C), 22.81 (-CH₃ 2C), 6.46 (Cyc-2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{15}\text{H}_{19}\text{N}_2\text{OS}_2$: 307.0905; found: 307.0917.

N-cyclobutyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-2):

Melting point: 108–111 °C. IR (KBr) cm^{-1} : 3286, 2924, 2854, 1725, 1641, 1550, 1449, 1403, 1380, 1359, 1314, 1266, 1228, 1165, 1127, 1100, 1073, 1034, 993, 902, 886, 864, 831, 795, 778, 722, 699, 672, 623. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.72 (d, J = 8.4 Hz, 1H, Ar-H), 8.4 (bs, 1H, -NH), 7.56 (d, J = 1.8 Hz, 1H, Ar-H), 7.28 (dd, J = 8.4 Hz, 1.2 Hz, 1H, Ar-H), 4.29–4.24 (m, 1H, Cyc-CH), 3.82 (s, 2H, -CH₂), 2.98–2.94 (m, 1H, -CH), 2.26–2.21 (m, 2H, Cyc-CH), 1.75–1.70 (m, 2H, Cyc-CH), 1.68–1.59 (m, 2H, Cyc-CH), 1.24 (d, J = 7.2 Hz, 6H, -2CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 167.3 (CO), 165.4 (Ar C), 150.7 (Ar C), 146.2 (Ar C), 135.7 (Ar C), 125.5 (Ar C), 120.9 (Ar C), 118.5 (Ar C), 45.0 (Cyc-C), 36.2 (-CH₂), 34.2 (-CH), 30.9 (Cyc-C), 24.2 (-CH₃ 2C), 15.2 (Cyc-2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{16}\text{H}_{21}\text{N}_2\text{OS}_2$: 321.1101; found: 321.1110.

N-cyclopentyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-3):

Melting point: 110–112 °C. IR (KBr) cm^{-1} : 3291, 2956, 2916, 2850, 1729, 1639, 1550, 1568, 1449, 1401, 1359, 1322, 263, 1225, 1194, 1178, 1154, 1098, 1073, 993, 903, 862, 830, 719, 672, 621. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.74 (d, J = 7.2 Hz, 1H, Ar-H), 7.64–7.62 (m, 2H, Ar-H, -NH), 7.32 (d, J = 7.2 Hz, 1H, Ar-H), 4.19–4.17 (m, 1H, -CH), 3.87 (s, 2H, -CH₂), 3.03–3.02 (m, 1H, -CH), 1.90 (m, 2H, Cyc-CH), 1.62–1.56 (m, 6H, Cyc-CH), 1.31 (s, 6H, -2CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 167.8 (CO), 165.5 (Ar C), 150.7 (Ar C), 146.1 (Ar C), 135.7 (Ar C), 125.5 (Ar C), 120.81 (Ar C), 118.5 (Ar C), 51.5 (Cyc-C), 36.8 (-CH₂), 34.2 (-CH), 32.9 (Cyc-2C), 24.2 (-CH₃ 2C), 23.6 (Cyc-2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{17}\text{H}_{25}\text{N}_2\text{OS}_2$: 322.1505; found: 322.2001.

N-cycloheptyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-4):

Melting point: 111–115 °C. IR (KBr) cm^{-1} : 3261, 2900, 2834, 2332, 2320, 1720, 1645, 1623, 1516, 1400, 1323, 1260, 1247, 1231, 1140, 1119, 1065, 1035, 1018, 970, 957, 919, 885, 870, 816, 820, 800, 752, 721, 700, 641, 615. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.69 (d, J = 8.4 Hz, 1H, -NH), 7.55 (d, J = 1.8 Hz, 1H, Ar-H), 7.46 (d, J = 6.6 Hz, 1H, Ar-H), 7.25 (dd, J = 8.4 Hz, 1.8 Hz, 1H, Ar-H), 3.90–3.85 (m, 1H, -CH), 3.82 (s, 2H, -CH₂), 2.98–2.94 (m, 1H, -CH, Cyc-CH), 1.79–1.76 (m, 2H, Cyc-CH), 1.53 (s, 3H, Cyc-CH), 1.47–1.45 (m, 3H, Cyc-CH), 1.39–1.33 (m, 4H, Cyc-CH), 1.74 (d, J = 8.4 Hz, 1H, Cyc-CH), 1.24 (d, J = 6.6 Hz, 6H, -2CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 167.1 (CO), 165.4 (Ar C), 150.8 (Ar C), 146.1 (Ar C), 135.7 (Ar C), 125.8 (Ar C), 120.9 (Ar C), 118.5 (Ar C), 50.6 (Cyc-C), 36.5 (-CH₂), 34.6 (-CH), 34.2 (Cyc 2C), 28.0 (Cyc 2C), 24.2 (-CH₃ 2C), 23.8 (Cyc 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{20}\text{H}_{27}\text{N}_2\text{OS}_2$: 364.1609; found: 364.16015.

N-(adamantan-1-yl)-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)

acetamide (C4-5):

Melting point: 125–130 °C. IR (KBr) cm^{-1} : 3939, 1897, 3276, 3017, 2957, 2901, 2847, 1674, 2352, 1656, 1551, 1443, 1411, 1361, 1237, 1209, 1122, 1101, 1072, 999, 950, 903, 884, 865, 822, 794, 739, 679, 640, 622. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.74 (d, J = 8.4 Hz, 1H, Ar—H), 7.61 (d, J = 1.8 Hz, 1H, Ar—H), 7.0–8.39 (bs, 1H, -NH), 7.31 (dd, J = 8.4 Hz, 1.8 Hz 1H, Ar—H), 3.80 (s, 2H, -CH₂), 3.04–3.00 (m, 1H, -CH), 2.02 (s, 3H, Cyc-CH), 1.95 (d, J = 6 Hz, 5H, Cyc-CH), 1.63 (s, 5H, -2CH₃), 1.30 (d, J = 7.2 Hz, 1H, -CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 167.2 (CO), 165.7 (Ar C), 150.7 (Ar C), 146.0 (Ar C), 135.7 (Ar C), 125.4 (Ar C), 120.8 (Ar C), 118.5 (Ar C), 51.9 (Cyc—C), 41.2 (Cyc 3C), 37.4 (Cyc 1C), 36.3 (-CH₂), 34.2 (-CH), 29.4 (Cyc 3C), 24.2 (-CH₃ 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{22}\text{H}_{29}\text{N}_2\text{O}_5\text{S}_2$: 401.1719; found: 401.1720.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-propylacetamide (C4-6):

Melting point: 115–122 °C. IR (KBr) cm^{-1} : 3293, 3086, 2959, 2929, 2869, 1640, 1554, 1467, 1449, 1404, 1381, 1360, 1326, 1305, 1266, 1243, 1226, 1157, 1129, 1099, 1073, 994, 952, 923, 902, 886, 862, 833, 779, 724, 673, 623. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.76 (d, J = 8.4 Hz, 1H, Ar—H), 7.62 (d, J = 1.2 Hz, 1H, Ar—H), 7.51 (bs, 1H, -NH), 7.32 (d, J = 8.4 Hz, 1.8 Hz, 1H, Ar—H), 3.93 (s, 2H, -CH₂), 3.24–3.20 (m, 2H, -CH₂), 3.05–3.00 (m, 1H, -CH), 1.52–1.50 (m, 2H, -CH₂), 1.30 (d, J = 7.2 Hz, 6H, -2 CH₃), 0.80 (d, J = 7.2 Hz, 3H, -CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 168.3 (CO), 165.3 (Ar C), 150.8 (Ar C), 146.1 (Ar C), 135.7 (Ar C), 125.5 (Ar C), 121.0 (Ar C), 118.5 (Ar C), 41.5 (-CH₂), 36.2 (-CH₂), 34.2 (-CH), 24.2 (-CH₃ 2C), 22.5 (-CH₂), 11.3 (-CH₃ C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{15}\text{H}_{21}\text{N}_2\text{O}_5\text{S}_2$: 309.1200; found: 309.1207.

N-butyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-7):

Melting point: 121–130 °C. IR (KBr) cm^{-1} : 3290, 3088, 2959, 2928, 2868, 1932, 1725, 1641, 1552, 1449, 1404, 1379, 1360, 1307, 1266, 1242, 1225, 1156, 1128, 1099, 1073, 994, 922, 902, 881, 863, 832, 778, 735, 689, 672, 623. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.69 (d, J = 8.4 Hz, 1H, Ar—H), 7.55 (d, J = 1.2 Hz, 1H, Ar—H), 7.39 (s, 1H, -NH), 7.25 (dd, J = 8.4 Hz, 1.8 Hz, 1H, Ar—H), 3.86 (s, 2H, -CH₂), 3.19–3.16 (m, 2H, -CH₂), 2.99–2.93 (m, 1H, -CH), 1.39–1.34 (m, 2H, -CH₂), 1.23 (d, J = 6.6 Hz, 6H, -2 CH₃), 1.20–1.17 (m, 2H, -CH₂), 0.75 (t, J = 1.8 Hz, 3H, -CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 168.2 (CO), 165.3 (Ar C), 150.9 (Ar C), 146.2 (Ar C), 135.7 (Ar C), 125.4 (Ar C), 121.0 (Ar C), 118.5 (Ar C), 39.5 (-CH₂), 36.2 (-CH₂), 34.2 (-CH), 31.2 (-CH₂), 24.2 (-CH₃ 2C), 20.0 (-CH₂), 13.6 (-CH₃ C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{16}\text{H}_{23}\text{N}_2\text{O}_5\text{S}_2$: 323.1319; found: 323.1315.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-pentylacetamide (C4-8):

Melting point: 125–130 °C. IR (KBr) cm^{-1} : 3346, 2956, 2921, 2855, 2356, 1884, 1726, 1657, 1598, 1551, 1516, 1443, 1411, 1375, 1332, 1311, 1244, 1212, 1178, 1146, 1116, 1070, 1000, 931, 906, 870, 819, 764, 740, 726, 676, 621. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.69 (d, J = 8.4 Hz, 1H, Ar—H), 7.54 (d, J = 1.8 Hz, 1H, Ar—H), 7.30 (s, 1H, -NH), 7.25 (dd, J = 8.4 Hz, 1.2 Hz, 1H, Ar—H), 3.86 (s, 2H, -CH₂), 3.18–3.15 (m, 2H, -CH₂), 2.98–2.93 (s, 1H, -CH), 1.39–1.35 (m, 2H, -CH₂), 1.23 (d, J = 6.6 Hz, 6H, -2 CH₃), 1.14–1.09 (m, 4H), 0.68 (t, J = 7.2 Hz, 3H); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 168.2 (CO), 165.3 (Ar C), 150.8 (Ar C), 146.2 (Ar C), 135.7 (Ar C), 125.4 (Ar C), 121.0 (Ar C), 118.3 (Ar C), 39.8 (-CH₂), 36.1 (-CH₂), 34.2 (-CH), 29.0 (-CH₂), 28.9 (-CH₂), 24.2 (-CH₃ 2C), 22.3 (-CH₂), 13.8 (-CH₃ C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{17}\text{H}_{25}\text{N}_2\text{O}_5\text{S}_2$: 337.1420; found: 337.1419.

N-hexyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-9):

Melting point: 128–132 °C. IR (KBr) cm^{-1} : 3018, 2949, 2918, 2878, 1912, 1700, 1621, 1512, 1429, 1404, 1339, 1320, 1246, 1222, 1136, 1118, 1089, 1053, 983, 900, 865, 802, 778, 745, 679, 652, 611. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.68 (d, J = 8.4 Hz, 1H, Ar—H), 7.53 (d, J = 0.6 Hz, 1H, Ar—H), 7.39 (s, 1H, -NH), 7.24 (dd, J = 8.4 Hz, 1.2 Hz, 1H, Ar—H), 3.90 (s, 2H, -CH₂), 3.18–3.14 (m, 2H, -CH₂), 3.00–2.92 (m, 1H, -CH), 1.38–1.33 (m, 2H, -CH₂), 1.22 (d, J = 7.2 Hz, 6H, -2 CH₃), 1.12–1.05 (m, 6H, -2CH₂), 0.69 (t, J = 6.8 Hz, 3H, -CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 168.2 (CO), 165.2 (Ar C), 150.8 (Ar C), 146.1 (Ar C), 135.7 (Ar C), 125.4 (Ar C), 121.0 (Ar C), 118.5 (Ar C), 39.8 (-CH₂),

36.1 (-CH₂), 34.2 (-CH), 31.4 (-CH₂), 29.1 (-CH₂), 26.5 (-CH₂), 24.2 (-CH₃ 2C), 22.4 (-CH₂), 13.9 (-CH₃). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{18}\text{H}_{27}\text{N}_2\text{O}_5\text{S}_2$: 351.1600; found: 351.1610.

N-isobutyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-10):

Melting point: 115–120 °C. IR (KBr) cm^{-1} : 3290, 2950, 2928, 2861, 1922, 1715, 1640, 1522, 1459, 1401, 1369, 1350, 1309, 1255, 1232, 1225, 1121, 1079, 1043, 990, 912, 900, 873, 822, 768, 678, 633. ^1H NMR (CDCl_3 , 500 MHz) δ = 7.76 (d, J = 8.5 Hz, 1H, Ar—H), 7.65–7.63 (m, 1H, Ar—H), 7.32 (d, J = 8.5 Hz, 1.5 Hz, 1H, Ar—H), 3.95 (s, 2H, -CH₂), 3.12–3.10 (m, 2H, -CH₂), 3.07–3.01 (m, 1H, -CH), 1.79–1.71 (m, 1H, -CH), 1.32 (d, J = 7 Hz, 6H, -2 CH₃), 0.87 (d, J = 6.5 Hz, 6H, 2 CH₃); ^{13}C NMR (CDCl_3 , 125 MHz) δ = 168.4 (CO), 165.5 (Ar C), 150.7 (Ar C), 146.1 (Ar C), 135.7 (Ar C), 125.5 (Ar C), 121.0 (Ar C), 118.5 (Ar C), 47.1 (-CH₂), 36.2 (-CH₂), 34.2 (-CH), 28.4 (-CH), 24.1 (-2CH₃), 20.0 (-CH₃). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{16}\text{H}_{23}\text{N}_2\text{O}_5\text{S}_2$: 323.1311; found: 322.1323.

N,N-diethyl-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-11):

Melting point: 118–120 °C. IR (KBr) cm^{-1} : 3058, 2945, 2908, 2858, 1912, 1717, 1601, 1532, 1432, 1404, 1359, 1342, 1301, 1256, 1222, 1166, 1118, 1095, 1070, 964, 919, 903, 885, 859, 830, 768, 725, 682, 669, 619. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.74 (d, J = 8.4 Hz, 1H, Ar—H), 7.6 (s, 1H, -NH), 7.28–7.26 (m, 1H, Ar—H), 4.37 (s, 2H, -CH₂), 3.50–3.46 (m, 2H, -CH₂), 3.45–3.41 (m, 2H, -CH₂), 3.03–2.98 (m, 1H, -CH), 1.29 (d, J = 7.2 Hz, 9H, (-3CH₃), 1.15 (t, J = 7.2 Hz, 3H, (-CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 166.2 (CO), 151.3 (Ar C), 145.6 (Ar C), 125.1 (Ar C), 121.0 (Ar C), 118.3 (Ar C), 42.7 (-CH₂), 40.8 (-CH₂), 36.7 (-CH), 34.2 (-CH), 24.2 (-CH₃ 2C), 14.4 (-2CH₃), 12.9 (-2CH₃). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{16}\text{H}_{23}\text{N}_2\text{O}_5\text{S}_2$: 323.1319; found: 322.1322.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-phenylacetamide (C4-12):

Melting point: 110–114 °C. IR (KBr) cm^{-1} : 3269, 3201, 3139, 3087, 2953, 2904, 2865, 2358, 1660, 1600, 1547, 1496, 1464, 1441, 1410, 1386, 1332, 1290, 1244, 1192, 1163, 1131, 1099, 1070, 1036, 1010, 951, 894, 875, 843, 813, 753, 719, 687, 659, 818. ^1H NMR (CDCl_3 , 500 MHz) δ = 10.2 (s, 1H, -NH), 7.90 (d, J = 8.5 Hz, 1H, Ar—H), 7.67 (d, J = 1.5 Hz, 1H, Ar—H), 7.55–7.54 (m, 2H, Ar—H), 7.41 (d, J = 8 Hz, 1.5 Hz, 1H, Ar—H), 7.32 (t, J = 8 Hz, 2H, Ar—H), 7.09 (t, J = 7.5 Hz, 1H, Ar—H), 4.07 (s, 2H, -CH₂), 3.10–3.05 (m, 1H, -CH), 1.35 (d, J = 7 Hz, 6H, -2 CH₃); ^{13}C NMR (CDCl_3 , 125 MHz) δ = 166.8 (CO), 166.4 (Ar C), 150.4 (Ar C), 146.4 (Ar C), 138.2 (Ar C), 135.7 (Ar C), 129.0 (Ar 2C), 125.8 (Ar C), 124.2 (Ar C), 120.7 (Ar 2C), 119.5 (Ar 2C), 118.7 (Ar C), 37.4 (-CH₂), 34.3 (-CH), 24.2 (-CH₃ 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{19}\text{H}_{23}\text{N}_2\text{O}_5\text{S}_2$: 359.1304; found: 359.1308.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-(4-methoxyphenyl)acetamide (C4-13):

Melting point: 145–150 °C. IR (KBr) cm^{-1} : 3200, 3157, 2824, 2220, 1689, 1523, 1459, 1388, 1350, 1324, 1262, 1242, 1224, 1149, 1120, 1068, 1021, 1005, 959, 920, 849, 816, 736, 700, 694, 949, 919, 619. ^1H NMR (CDCl_3 , 600 MHz) δ = 9.93 (s, 1H, -NH), 7.85 (d, J = 8.4 Hz, 1H, Ar—H), 7.63 (s, 1H, Ar—H), 7.42 (d, J = 9 Hz, 2H, Ar—H), 7.36 (d, J = 8.4 Hz, 1H, Ar—H), 6.82 (d, J = 8.4 Hz, 2H, Ar—H), 4.03 (s, 2H, -CH₂), 3.75 (s, 3H, -OCH₃), 3.06–3.01 (m, 1H, -CH), 1.31 (d, J = 6.6 Hz, 6H, -2 CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 166.4 (CO), 166.3 (Ar C), 156.3 (Ar C), 150.4 (Ar C), 146.4 (Ar C), 135.7 (Ar C), 131.4 (Ar C), 125.7 (Ar C), 121.1 (Ar 2C), 120.7 (Ar C), 118.7 (Ar C), 114.2 (Ar 2C), 55.5 (-OCH₃), 37.2 (-CH₂), 34.2 (-CH), 24.2 (-CH₃ 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{19}\text{H}_{21}\text{N}_2\text{O}_5\text{S}_2$: 373.1008; found: 373.1018.

N-(4-fluorophenyl)-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-14):

Melting point: 110–115 °C. IR (KBr) cm^{-1} : 3259, 3201, 3180, 3129, 3057, 2849, 2921, 2853, 2347, 2330, 1681, 1580, 1525, 1489, 1400, 1380, 1358, 1321, 1301, 1245, 1162, 1115, 1051, 1009, 948, 907, 889, 835, 737, 671, 656, 615. ^1H NMR (CDCl_3 , 600 MHz) δ = 10.21 (s, 1H, -NH), 7.87 (d, J = 8.5 Hz, 1H, Ar—H), 7.67 (d, J = 1.5 Hz, 1H, Ar—H),

7.51–7.49 (m, 2H, Ar–H), 7.41 (dd, $J = 8.5$ Hz, 1.5 Hz, 1H, Ar–H), 7.02–6.99 (m, 2H, Ar–H), 4.06 (s, 2H, $-\text{CH}_2-$), 4.10–3.05 (m, 1H, $-\text{CH}$), 1.34 (d, $J = 7$ Hz, 6H, -2CH_3); ^{13}C NMR (CDCl_3 , 125 MHz) $\delta = 166.7$ (CO), 166.5 (Ar C), 159.2 (d, $J_{\text{F-C}} = 241.75$ Hz) (Ar C), 150.3 (Ar C), 146.5 (Ar C), 135.7 (Ar C), 134.3 (Ar C), 125.8 (Ar C), 121.1 (d, $J_{\text{F-C}} = 7.75$ Hz) (Ar C), 120.6 (d, $J_{\text{F-C}} = 9.8$ Hz) (Ar C), 118.8 (Ar 2C), 118.7 (Ar C), 115.7 (Ar 2C), 37.2 ($-\text{CH}_2-$), 34.3 ($-\text{CH}$), 24.1 ($-\text{CH}_3$ 2C); ^{19}F NMR (471 MHz, CDCl_3) $\delta = -111.1$ (m, 1F). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{18}\text{H}_{18}\text{FN}_2\text{O}_5$: 361.1010; found: 361.1017.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-(4-(trifluoromethyl)phenyl)acetamide (C4-15).

Melting point: 112–115 °C. IR (KBr) cm^{-1} : 3269, 2303, 3131, 3068, 2941, 2863, 1688, 1611, 1559, 1445, 1409, 1330, 1274, 1159, 1137, 1104, 1065, 1001, 850, 822, 713. ^1H NMR (CDCl_3 , 600 MHz) $\delta = 10.56$ (s, 1H, $-\text{NH}$), 7.87 (d, $J = 8.4$ Hz, 1H, Ar–H), 7.66–7.63 (m, 3H, Ar–H), 7.54 (d, $J = 8.4$ Hz, 2H, Ar–H), 7.40 (dd, $J = 8.4$ Hz, 1.2 Hz, 1H, Ar–H), 4.05 (s, 2H, $-\text{CH}_2-$), 3.08–3.04 (m, 1H, $-\text{CH}$), 1.33 (d, $J = 6.66$ Hz, $J = 6\text{H}$, -2CH_3); ^{13}C NMR (CDCl_3 , 150 MHz) $\delta = 167.2$ (CO), 166.6 (Ar C), 150.2 (Ar C), 146.7 (Ar C), 141.3 (Ar C), 135.7 (Ar C), 126.7 (d, $J_{\text{F-C}} = 3.6$ Hz) (Ar C), 125.9 (Ar 2C), 120.5 (Ar C), 119.1 (Ar C), 118.8 (Ar 2C), 37.4 ($-\text{CH}_2-$), 34.3 ($-\text{CH}$), 24.2 ($-\text{CH}_3$ 2C); ^{19}F NMR (471 MHz, CDCl_3) $\delta = -61.36$ (s, 3F). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{19}\text{H}_{18}\text{F}_3\text{N}_2\text{O}_5$: 411.0800; found: 411.1008.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-(4-(trifluoromethoxy)phenyl)acetamide (C4-16):

Melting point: 100–102 °C. IR (KBr) cm^{-1} : 3276, 2960, 2924, 2363, 2358, 1663, 1610, 1548, 1506, 1464, 1442, 1410, 1388, 1333, 1243, 1217, 1199, 1160, 1107, 1071, 1010, 952, 919, 895, 847, 821, 800, 765, 737, 706, 661. ^1H NMR (CDCl_3 , 500 MHz) $\delta = 10.41$ (s, 1H, $-\text{NH}$), 7.88 (d, $J = 8.5$ Hz, 1H, Ar–H), 7.68 (d, $J = 1.5$ Hz, 1H, Ar–H), 7.58–7.55 (m, 2H, Ar–H), 7.42 (dd, $J = 8.5$ Hz, 2 Hz, 1H, Ar–H), 7.18 (d, $J = 8.5$ Hz, 2H, Ar–H), 4.06 (s, 2H, $-\text{CH}_2-$), 3.11–3.05 (m, 1H, $-\text{CH}$), 1.35 (d, $J = 7$ Hz, 6H, -2CH_3); ^{13}C NMR (CDCl_3 , 125 MHz) $\delta = 166.9$ (CO), 166.6 (Ar C), 150.2 (Ar C), 146.6 (Ar C), 145.1 (Ar C), 137.0 (Ar C), 135.7 (Ar C), 125.9 (Ar C), 121.8 (Ar 2C), 121.5 (Ar C), 120.6 (Ar C), 120.5 (d, $J_{\text{F-C}} = 4.6$ Hz) (Ar C), 119.4 (Ar C), 118.8 (Ar 2C), 37.3, 34.3 ($-\text{CH}$), 24.1 ($-\text{CH}_3$), 24.0 ($-\text{CH}_3$). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{19}\text{H}_{18}\text{F}_3\text{N}_2\text{O}_5$: 427.1015; found: 427.1025.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-(naphthalen-2-yl)acetamide (C4-17):

Melting point: 108–110 °C. IR (KBr) cm^{-1} : 3321, 2051, 2995, 2945, 2917, 2859, 2362, 1904, 1793, 1686, 1627, 1547, 1498, 1469, 1435, 1402, 1336, 1262, 1208, 1130, 1070, 1006, 909, 874, 785, 729, 643, 610. ^1H NMR (CDCl_3 , 500 MHz) $\delta = 10.09$ (s, 1H, $-\text{NH}$), 8.10 (d, $J = 7.5$ Hz, 1H, Ar–H), 7.96 (d, $J = 8.5$ Hz, 1H, Ar–H), 7.88 (d, $J = 8.5$ Hz, 1H, Ar–H), 7.80 (d, $J = 8.5$ Hz, 1H, Ar–H), 7.63–7.62 (m, 2H, Ar–H), 7.46–7.40 (m, 2H, Ar–H), 7.34–7.30 (m, 2H, Ar–H), 4.21 (s, 2H, $-\text{CH}_2-$), 3.06–2.97 (m, 1H, $-\text{CH}$), 1.28 (d, $J = 7$ Hz, 6H, -2CH_3); ^{13}C NMR (CDCl_3 , 125 MHz) $\delta = 167.4$ (CO), 166.4 (Ar C), 150.5 (Ar C), 146.4 (Ar C), 135.8 (Ar C), 134.0 (Ar C), 132.8 (Ar C), 128.6 (Ar C), 126.5 (Ar C), 126.1 (Ar C), 125.9 (Ar C), 125.6 (Ar C), 125.2 (Ar C), 121.1 (Ar C), 121.1 (Ar C), 120.9 (Ar C), 119.5 (Ar C), 118.6 (Ar C), 37.3 ($-\text{CH}_2-$), 34.2 ($-\text{CH}$), 24.1 ($-\text{CH}_3$ 2C). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{22}\text{H}_{21}\text{N}_2\text{O}_5$: 393.1111; found: 393.1119.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-methyl-N-phenylacetamide (C4-18):

Melting point: 110–115 °C. IR (KBr) cm^{-1} : 2960, 2925, 2875, 2356, 1726, 1662, 1590, 1490, 1442, 1410, 1367, 1333, 1296, 1238, 1111, 1071, 1045, 1002, 921, 874, 817, 778, 740, 698, 670, 850, 618. ^1H NMR (CDCl_3 , 600 MHz) $\delta = 7.69$ (d, $J = 8.4$ Hz, 1H, Ar–H), 7.56 (s, 1H, $-\text{NH}$), 7.45 (t, $J = 7.2$ Hz, 2H, Ar–H), 7.37 (t, $J = 7.2$ Hz, 1H, Ar–H), 7.32 (d, $J = 8.5$ Hz, 2H, Ar–H), 7.25 (s, 1H, Ar–H), 4.03 (s, 2H, $-\text{CH}_2-$), 3.33 (s, 3H, $-\text{CH}_3$), 3.02–2.97 (m, 1H, $-\text{CH}$), 1.28 (d, $J = 7.3$ Hz, 6H, -2CH_3); ^{13}C NMR (CDCl_3 , 150 MHz) $\delta = 167.2$ (CO), 164.4 (Ar C), 151.3 (Ar C), 145.5 (Ar C), 143.3 (Ar C), 135.7 (Ar C), 130.0 (Ar C), 128.4 (Ar 2C), 127.3 (Ar 2C), 125.0 (Ar C), 121.1 (Ar C), 118.2 (Ar C), 38.0 (1 $-\text{CH}_2-$),

36.7 ($-\text{CH}_3$), 34.2 ($-\text{CH}$), 24.2 ($-\text{CH}_3$ 2C). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{20}\text{H}_{25}\text{N}_2\text{O}_5$: 373.1410; found: 373.1421.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-(thiazol-2-yl)acetamide (C4-19):

Melting point: 108–112 °C. IR (KBr) cm^{-1} : 2950, 2910, 2856, 1821, 1780, 1430, 1419, 1354, 1320, 1281, 1256, 1163, 1103, 1042, 1025, 954, 807, 670. ^1H NMR (CDCl_3 , 600 MHz) $\delta = 7.95$ (d, $J = 8.4$ Hz, 1H, Ar–H), 7.62 (s, 1H, $-\text{NH}$), 7.74 (d, $J = 3$ Hz, 1H, Ar–H), 7.37 (dd, $J = 8.4$ Hz, 1.2 Hz, 1H, Ar–H), 6.98 (d, $J = 3.6$ Hz, 1H, Ar–H), 4.16 (s, 2H, $-\text{CH}_2-$), 3.06–3.01 (m, 1H, $-\text{CH}$), 1.30 (d, $J = 6.6$ Hz, 6H, -2CH_3); ^{13}C NMR (CDCl_3 , 150 MHz) $\delta = 166.3$ (CO), 165.3 (Ar C), 157.9 (Ar C), 150.3 (Ar C), 146.5 (Ar C), 137.8 (Ar C), 135.5 (Ar C), 125.8 (Ar C), 121.3 (Ar C), 118.4 (Ar C), 113.8 (Ar C), 36.1 ($-\text{CH}_2-$), 34.2 ($-\text{CH}$), 24.1 ($-\text{CH}_3$ 2C). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{15}\text{H}_{16}\text{N}_3\text{O}_5$: 350.0616; found: 350.1011.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-(pyridin-2-yl)acetamide (C4-20):

Melting point: 130–138 °C. IR (KBr) cm^{-1} : 3184, 3108, 3040, 2958, 2923, 2858, 2355, 2338, 1727, 1691, 1581, 1533, 1457, 1435, 1410, 1379, 1359, 1313, 1290, 1267, 1236, 1159, 1132, 1100, 1069, 1037, 1003, 959, 903, 869, 770, 732, 663, 620. ^1H NMR (CDCl_3 , 500 MHz) $\delta = 10.27$ (s, 1H, $-\text{NH}$), 8.29–8.28 (m, 1H, Ar–H), 8.22 (d, $J = 8.5$ Hz, 1H, Ar–H), 7.96 (d, $J = 8.5$ Hz, 1H, Ar–H), 7.71–7.67 (m, 1H, Ar–H), 7.62 (d, $J = 1.5$ Hz, 1H, Ar–H), 7.36 (dd, $J = 8.5$ Hz, 1.5 Hz, 1H, Ar–H), 7.03–7.01 (m, 1H, Ar–H), 4.15 (m, 2H, $-\text{CH}_2-$), 3.07–3.01 (m, 1H, $-\text{CH}$), 1.32 (d, $J = 7$ Hz, 6H, -2CH_3); ^{13}C NMR (CDCl_3 , 125 MHz) $\delta = 167.1$ (CO), 164.9 (Ar C), 151.4 (Ar C), 150.7 (Ar C), 148.2 (Ar C), 146.2 (Ar C), 138.2 (Ar C), 138.1 (Ar C), 135.7 (Ar C), 125.5 (Ar 2C), 121.5 (Ar C), 119.8 (Ar C), 118.4 (Ar C), 114.1 (Ar C), 37.4 ($-\text{CH}_2-$), 34.3 ($-\text{CH}$), 24.2 ($-\text{CH}_3$ 2C). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{18}\text{H}_{22}\text{N}_3\text{O}_5$: 360.5101; found: 360.5113.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-(5-methoxypyridin-2-yl)acetamide (C4-21):

Melting point: 156–160 °C. IR (KBr) cm^{-1} : 2957, 2357, 2324, 2020, 1679, 1587, 1537, 1449, 1408, 1390, 1314, 1282, 1252, 1224, 1169, 1140, 1068, 1032, 1000, 964, 921, 859, 826, 746, 721, 694, 959, 929, 620. ^1H NMR (CDCl_3 , 600 MHz) $\delta = 10.10$ (s, 1H, $-\text{NH}$), 8.11 (d, $J = 9$ Hz, 1H, Ar–H), 7.95 (d, $J = 3$ Hz, 1H, Ar–H), 7.93 (d, $J = 8.4$ Hz, 1H, Ar–H), 7.60 (s, 1H, Ar–H), 7.34 (d, $J = 8.4$ Hz, 1H, Ar–H), 7.22 (dd, $J = 9$ Hz, 3 Hz, 1H, Ar–H), 4.12 (s, 2H, $-\text{CH}_2-$), 3.81 (s, 3H, $-\text{OCH}_3$), 3.04–3.00 (m, 1H, $-\text{CH}$), 1.30 (d, $J = 7.2$ Hz, 6H, -2CH_3); ^{13}C NMR (CDCl_3 , 150 MHz) $\delta = 166.5$ (CO), 164.9 (Ar C), 152.8 (Ar C), 150.7 (Ar C), 146.2 (Ar C), 145.1 (Ar C), 135.7 (Ar C), 134.7 (Ar C), 125.5 (Ar C), 123.09 (Ar C), 121.4 (Ar C), 118.4 (Ar C), 114.5 (Ar C), 55.9 ($-\text{OCH}_3$), 37.2 ($-\text{CH}_2-$), 34.2 ($-\text{CH}$), 24.2 ($-\text{CH}_3$ 2C). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{18}\text{H}_{20}\text{N}_3\text{O}_5$: 374.1012; found: 374.1011.

N-(5-fluoropyridin-2-yl)-2-((6-isopropylbenzo[d]thiazol-2-yl)thio)acetamide (C4-22):

Melting point: 107–110 °C. IR (KBr) cm^{-1} : 3251, 3221, 3190, 3119, 3061, 2956, 2922, 2862, 2358, 2338, 1693, 1599, 1536, 1499, 1408, 1387, 1358, 1331, 1308, 1240, 1160, 1125, 1071, 1004, 968, 903, 869, 845, 827, 747, 690, 666, 625. ^1H NMR (CDCl_3 , 500 MHz) $\delta = 10.46$ (s, 1H, $-\text{NH}$), 8.24–8.21 (m, 1H, Ar–H), 8.22 (d, $J = 3.6$ Hz, 1H, Ar–H), 7.96 (d, $J = 10.2$ Hz, 1H, Ar–H), 7.64 (d, $J = 1.8$ Hz, 1H, Ar–H), 7.45–7.41 (m, 1H, Ar–H), 7.38 (dd, $J = 9.6$ Hz, 1.8 Hz, 1H, Ar–H), 4.12 (s, 2H, $-\text{CH}_2-$), 3.06–3.03 (m, 1H, $-\text{CH}$), 1.33 (d, $J = 7.8$ Hz, 6H, -2CH_3); ^{13}C NMR (CDCl_3 , 125 MHz) $\delta = 166.9$ (CO), 165.1 (Ar C), 156.4 (d, $J_{\text{F-C}} = 250$ Hz) (Ar C), 150.6 (Ar C), 147.4 (d, $J_{\text{F-C}} = 2.25$ Hz) (Ar C), 146.3 (Ar C), 136.7 (d, $J_{\text{F-C}} = 3.62$ Hz) (Ar C), 135.53 (d, $J_{\text{F-C}} = 10.2$ Hz) (Ar C), 125.62 (Ar C), 124.9 (d, $J_{\text{F-C}} = 19.2$ Hz) (Ar C), 121.45 (Ar C), 121.37 (Ar C), 118.49 (d, $J_{\text{F-C}} = 8.25$ Hz) (Ar C), 114.83, 37.3 ($-\text{CH}_2-$), 34.3 ($-\text{CH}$), 24.2 ($-\text{CH}_3$ 2C), 24.1 ($-\text{CH}_3$ 2C). ^{19}F NMR (471 MHz, CDCl_3) $\delta = -111.2$ (m, 1F). HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd. for $\text{C}_{17}\text{H}_{17}\text{FN}_3\text{O}_5$: 362.0811; found: 362.0819.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-N-(pyridin-2-ylmethyl)acetamide (C4-23):

Melting point: 115–120 °C. IR (KBr) cm^{-1} : 3290, 3054, 2959, 2952, 2868, 1659, 1592, 1569, 1525, 1460, 1440, 1411, 1382, 1363, 1306, 1242, 1207, 1149, 1101, 1073, 1049, 1002, 903, 868, 822, 753, 672, 623. ^1H NMR (CDCl_3 , 500 MHz) δ = 8.40 (d, J = 3 Hz, 1H, Ar—H), 8.40 (s, 1H, —NH), 7.78 (d, J = 8.5 Hz, 1H, Ar—H), 7.60 (d, J = 1 Hz, 1H, Ar—H), 7.54 (td, J = 8.4 Hz, 1H, Ar—H), 7.31 (dd, J = 8.5 Hz, 1.5 Hz, 1H, Ar—H), 7.19 (d, J = 8 Hz, 1H, Ar—H), 7.12–0.10 (m, 1H, Ar—H), 4.58 (d, J = 5.5 Hz, 2H, —CH₂), 4.08 (s, 2H, —CH₂), 3.05–3.00 (m, 1H, —CH), 1.30 (d, J = 7 Hz, 6H, —2 CH₃); ^{13}C NMR (CDCl_3 , 125 MHz) δ = 166.3 (CO), 164.7 (Ar C), 156.3 (Ar C), 150.9 (Ar C), 149.0 (Ar C), 146.0 (Ar C), 136.7 (Ar C), 125.7 (Ar C), 125.3 (Ar C), 122.2 (Ar C), 121.7 (Ar C), 121.4 (Ar C), 118.4 (Ar C), 45.1 (—CH₂), 36.1 (—CH₂), 34.1 (—CH), 24.4 (—CH₃ 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{19}\text{H}_{24}\text{N}_3\text{O}_5$: 374.5405; found: 374.5419.

2-((6-isopropylbenzo[d]thiazol-2-yl)thio)-N-(quinolin-5-yl)acetamide (C4-25):

Melting point: 140–145 °C. IR (KBr) cm^{-1} : 3962, 3915, 3880, 3813, 3336, 3294, 2952, 2356, 1677, 1594, 1594, 1548, 1492, 1465, 1444, 1407, 1362, 1325, 1286, 1263, 1241, 1202, 1133, 1076, 1060, 1001, 953, 902, 865, 822, 800, 780, 759, 739, 681, 655, 627. ^1H NMR (CDCl_3 , 600 MHz) δ = 10.22 (s, 1H, —NH), 8.88 (d, J = 3 Hz, 1H, Ar—H), 8.32 (d, J = 8.4 Hz, 1H, Ar—H), 8.12 (d, J = 7.2 Hz, 1H, Ar—H), 7.93 (d, J = 8.4 Hz, 1H, Ar—H), 7.84 (d, J = 8.4 Hz, 1H, Ar—H), 7.71 (d, J = 8.4 Hz, 1H, Ar—H), 7.67 (s, 1H, Ar—H), 7.37 (d, J = 8.4 Hz, 1H, Ar—H), 7.25–7.39 (m, 1H, Ar—H), 4.21 (s, 2H, —CH₂), 3.07–3.03 (m, 1H, —CH), 1.31 (d, J = 7.2 Hz, 6H, —2 CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 167.6 (CO), 166.6 (Ar C), 150.4 (Ar C), 150.2 (Ar C), 148.5 (Ar C), 146.7 (Ar C), 135.8 (Ar C), 132.9 (Ar C), 130.0 (Ar C), 129.5 (Ar C), 126.7 (Ar C), 125.8 (Ar C), 122.1 (Ar C), 120.8 (Ar C), 120.8 (Ar C), 120.3 (Ar C), 118.8 (Ar C), 37.3 (—CH₂), 34.2 (—CH), 24.2 (—CH₃ 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{21}\text{H}_{20}\text{N}_3\text{O}_5$: 394.5319; found: 394.5323.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-1-(piperidin-1-yl)ethan-1-one (C4-26):

IR (KBr) cm^{-1} : 2929, 2856, 1637, 1440, 1410, 1364, 1333, 1268, 1242, 1211, 1162, 1134, 1072, 1000, 952, 903, 867, 852, 821, 780, 742, 672, 623. ^1H NMR (CDCl_3 , 500 MHz) δ = 7.76 (d, J = 8.5 Hz, 1H, Ar—H), 7.59 (d, J = 1 Hz, 1H, Ar—H), 7.29–7.27 (m, 1H, Ar—H), 4.40 (s, 2H, —CH₂), 3.60–3.54 (m, 4H, —2CH₂), 3.04–2.98 (m, 1H, —CH), 1.69–1.63 (m, 4H, —2CH₂), 1.59–1.55 (m, 2H, —CH₂), 1.29 (d, J = 7 Hz, 6H, —2 CH₃); ^{13}C NMR (CDCl_3 , 125 MHz) δ = 165.3 (CO), 164.8 (Ar C), 151.3 (Ar C), 145.6 (Ar C), 135.6, 125.0 (Ar C), 121.0 (Ar C), 120.9 (Ar C), 47.5 (Cyc. Ring C), 43.4 (Cyc. Ring C), 36.7 (—CH₂), 34.1 (—CH), 26.5 (Cyc. Ring C), 25.5 (Cyc. Ring C), 24.2 (—CH₃ 2C), 24.1 (—Cyc. Ring C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{17}\text{H}_{23}\text{N}_2\text{O}_5$: 335.1809; found: 335.1819.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-1-morpholinoethan-1-one (C4-27):

IR (KBr) cm^{-1} : 2932, 2857, 1637, 1440, 1410, 1364, 1333, 1268, 1242, 1211, 1134, 1072, 1000, 952, 868, 821, 780, 742, 672, 623. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.74 (d, J = 8.4 Hz, 1H, Ar—H), 7.60 (d, J = 0.6 Hz, 1H, Ar—H), 7.28 (dd, J = 8.4 Hz, 1.2 Hz, 1H, Ar—H), 4.35 (s, 2H, —CH₂), 3.73–3.71 (m, 2H, —CH₂), 3.69–3.66 (m, 6H, —3CH₂), 3.04–2.97 (m, 1H, —CH), 1.29 (d, J = 7.2 Hz, 6H, —2 CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 166.1 (CO), 164.2 (Ar C), 151.2 (Ar C), 145.8 (Ar C), 135.7 (Ar C), 125.2 (Ar C), 121.0 (Ar C), 118.4 (Ar C), 66.7 (Cyc. Ring C), 66.6 (Cyc. Ring C), 46.8 (Cyc. Ring C), 42.6 (Cyc. Ring C), 35.5 (—CH₂), 34.2 (—CH), 24.2 (—CH₃ 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{16}\text{H}_{21}\text{N}_2\text{O}_5$: 337.1208; found: 337.1519.

2-((6-Isopropylbenzo[d]thiazol-2-yl)thio)-1-thiomorpholinoethan-1-one (C4-28):

IR (KBr) cm^{-1} : 2957, 2921, 2866, 1724, 1640, 1440, 1412, 1364, 1333, 1291, 1245, 1211, 1183, 1123, 1072, 1035, 1001, 954, 904, 868, 821, 780, 742, 670, 623. ^1H NMR (CDCl_3 , 600 MHz) δ = 7.76 (d, J = 8.4 Hz, 1H, Ar—H), 7.59 (s, 1H, Ar—H), 7.29 (dd, J = 8.4 Hz, 1.8 Hz, 1H, Ar—H), 4.35 (s, 2H, —CH₂), 3.91 (d, J = 4.8 Hz, 4H), 3.03–2.99 (m, 1H, —CH), 2.72 (t, J = 4.8 Hz, 2H), 2.64 (d, J = 5.4 Hz, 2H), 1.94 (d, J = 6.6 Hz, 6H, —2 CH₃); ^{13}C NMR (CDCl_3 , 150 MHz) δ = 165.9 (CO), 164.2 (Ar

C), 151.2 (Ar C), 145.8 (Ar C), 135.7 (Ar C), 125.2 (Ar C), 121.0 (Ar C), 118.4 (Ar C), 49.2 (Cyc. Ring C), 45.1 (Cyc. Ring C), 36.0 (—CH₂), 34.2 (—CH), 27.9 (Cyc. Ring C), 27.3 (Cyc. Ring C), 24.2 (—CH₃ 2C). HRMS (ESI-TOF) m/z : [M + H]⁺ Calcd. for $\text{C}_{16}\text{H}_{20}\text{N}_2\text{O}_5\text{S}_2$: 352.5314; found: 352.5228.

7. Biological

7.1. Mycobacterial growth conditions

To find the minimum inhibitory concentration (MIC) against derivative BSL2 strains of *Mtb* H37Rv, such as *Mtb* mc²6230, *Mtb* mc²8243 (resistant to isoniazid), *Mtb* mc²8247 (resistant to rifampicin), and *Mtb* mc²8259 (MDR-both resistant to isoniazid and rifampicin), the synthesized compounds were examined *in-vitro*. At 37 °C with 5 % CO₂, the bacteria were cultivated in Middle Brook (BD Difco laboratories) 7H9 liquid broth medium supplemented with 0.2 % glycerol, 0.05 % Tween 80, and 10 % Album-Dextrose-Saline (ADS), additionally supplemented with L-methionine (50 mg/L), L-leucine (50 mg/L), L-arginine (200 mg/L), and D-pantothenic acid (24 mg/L).

7.2. In-vitro determination of minimum inhibitory concentration (MIC)

The liquid broth micro-dilution method was used in 96-well round bottom microtiter plates (Corning, New York, USA) in order to determine the MIC of the synthesized compounds [6,9]. In short, the bacterial cultures were grown to 1.0×10^8 CFU/mL, or mid-log phase (i.e., OD_{600nm} range of 0.4–0.6). After diluting the final inoculum to an OD₆₀₀ of 0.005 (1.0×10^6 CFU/mL), it was prepared. Each well contained 100 μL of media that had been serially diluted twice for the test molecules. Lastly, 100 μL of mycobacterial strains were added to each plate well, and the plates were kept for 10 days at 37 °C. Following the incubation period, the plates were visually inspected, and the MIC—the lowest concentration of the molecule that showed no turbidity—was determined. **CBR-5992** and isoniazid were maintained as the standard positive controls in this assay.

7.3. Determination of minimum bactericidal concentration (MBC)

The bacterial inoculum was obtained from the mid log phase (i.e., OD 600 nm range of 0.4–0.6) and adjusted at an OD_{600nm} of 0.005 in accordance with in order to calculate the MBC₉₀ of the compounds [38,49]. For five days, the bacteria were exposed to various chemical concentrations at 37 °C. To measure the colony forming units (CFU), the bacteria were serially diluted ten times and spotted on 7H10 agar plates after five days of incubation. The agar plates were then kept for three to four weeks. MBC₉₀ was defined as a 90 % decrease in the bacterial population relative to the original inoculum. **BDQ** and **CBR-5992** were employed as positive controls.

7.4. Determination of intracellular ATP depletion

The amount of ATP present in the bacteria was measured using the BacTiter-Glo™ Microbial Cell Viability Assay (Promega, Madison, USA) in accordance with [36,46,47]. In essence, the bacteria were taken out of the mid-log phase and set to a 0.05 at OD:600 nm. In 96-well flat bottom white opaque microtiter plates (Corning, New York, USA), 50 μL of cultures were distributed after the compounds were added to each well at the appropriate concentration. For twenty-four hours, the plates were incubated at 37 °C. Additionally, each well received 25 μL of BacTiter-Glo reagent, which was then incubated for 12 min. The BioTek Cytation 5 Multiple-mode reader was used to take the luminescence reading, and GraphPad Prism 8 software was used to analyze the results. **BDQ** and **CBR-5992** were used as a standard drug in this assay.

7.5. Fluorimetric based determination of NADH/NAD⁺ ratio

The pMV762-peredox-mcherry-transformed *M. tuberculosis* strain was used in the NADH/NAD⁺ determination test. The *M. tuberculosis*-peredox-mCherry strain was cultivated in a log phase (OD_{600nm} 0.4–0.6) using appropriate 7H9 medium supplemented with necessary nutrients. In a 96-well black flat bottom with a clear bottom, 200 μ l of the bacterial suspension was directly inoculated per well after the compounds were dispensed into each well according to the appropriate concentration (Corning, New York, USA). The plates were incubated for 24 h at 37 °C, and the BioTek Cytation 5 Multiple-mode reader was used to measure the spectra. The excitation wavelengths were 400 nm and 587 nm, and the emission wavelengths were 510 nm and 615 nm, respectively [38,48,49,50]. The NADH/NAD⁺ ratio in the bacterial cells was calculated by plotting the ratio (green/red) using GraphPad Prism 8. **CBR-5992**, Clofazimine and Rotenone was used as a standard in this assay.

7.6. Determination of cytotoxicity using HepG2 cell lines

Using human hepatocellular carcinoma HepG2 cell lines (ATCC HB-8065) in DMEM glucose supplemented with 10 % foetal bovine serum (FBS), streptomycin (100 μ g/ml), penicillin (75 μ g/ml), HEPES (10 mM), and L-glutamine (2 mM) at 37 °C, the MTT [3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide] assay was used to assess the compounds' cell cytotoxicity. At a density of 1×10^4 cells/well, cells were seeded in a 96-well flat bottom (Tarsons) and cultured for 24 h at 37 °C with 5 % CO₂. The test compounds were serially diluted ten times, with a range of 1000–0.48 μ g/ml. Cell viability was assessed using a BioTek Cytation 5 Cell Imaging multiple-mode reader by measuring the absorbance of the reduced formazan dissolved in DMSO at 570 nm after MTT was administered at a concentration of 2.5 mg/ml dissolved in phosphate buffer saline. GraphPad Prism 8 was used to analyze the data. Cytotoxicity is expressed as CC₅₀, the concentration that results in a 50 % decrease in cell viability. In this experiment, tamoxifen was used as a standard [11].

7.7. Determination of selectivity index (SI)

The ratio of a compound's cytotoxicity (CC₅₀) to biological activity (MIC) determines the test chemicals' selectivity index. SI must be greater than 10 for a chemical to be in the therapeutic window [11].

7.8. Target confirmation through one-step mutant generation

By plating 10⁹ CFU/mL of *Mtb* mc²6230 on 7H10 agar plates supplemented with 10 % OADC and NDH-2 inhibitor at a concentration corresponding to the 8 times MIC value, the one-step mutants for a known NDH-2 inhibitor, i.e., **CBR-5992**, were chosen. The colonies were selected and cultured to log phase in 7H9 liquid broth supplemented with 10 % ADC after the plates were incubated for three weeks at 37 °C. Using a liquid broth microdilution experiment, the susceptibilities of these mutants were assessed for **CBR-5992**. Additionally, they were examined for any cross-resistance to the test active molecules. Mutants that demonstrated resistance to **CBR-5992** that was greater than eight times the MIC values and SNPs in the NDH-2 encoding gene were found by processing test active molecules in relation to the susceptible strain [36,42].

7.9. Molecular docking study

The structure of type II NADH: quinone oxidoreductase with 2-mercapto-quinazolinone inhibitor was used for docking (PDB ID: 9MQZ) [51]. The structural refinement of the ligands was done using auto dock 1.5.7 prior to docking. Docking was performed using AutoDock Vina 1.1.2. Input is provided in the form of receptor, ligand, and a

configuration file consisting of information on the size and centers of the grid box, while output is in the form of a list of poses ranked by a score that is corresponding to the predicted binding energy (ΔG , kcal/mol). The exhaustiveness was set at 8, the number of binding modes was set at 9, and the energy range of all the binding poses was fixed at 3 kcal/mol. Docking of the molecules (**CBR-5992**, **C4**, **C4-1**, and **C4-11**) against *Mtb* Ndh-2 protein was done by rigidifying the search place at the 2-mercapto-quinazolinone binding site. All of the bound small molecules were manually removed before docking. After the ligands were docked against the receptor proteins, the best docked poses of all the molecules and the interactions with the binding residues of the proteins were visualized using BIOVIA Discovery Studio Visualizer v21.1.0.20298.

7.10. Molecular dynamics (MD) simulation analysis

MD simulation was performed using GROMACS software, where gromacs- 2020 version 2020 was used. Input files were generated using CHARMM- GUI web server's solution builder program, where the system was built by providing the protein-ligand complex in protein database (.pdb) format. A rectangular box of water was generated using TIP3 model water, and the system was neutralised using Na⁺ and Cl[−] ions. The ions were added at a concentration of 0.15 M using the Monte Carlo method of ion placing. Once the system was generated, the temperature of 311.15 K and pressure of 1 bar were set, and the system was equilibrated using the energy minimisation step. After the minimum energy levels were obtained for the system, NVT (constant number of particles, volume and temperature) and NPT (constant number of particles, pressure and temperature) ensembles were performed following MD simulations for 100 ns (5×10^7 steps).

7.11. In silico physicochemical properties and ADME profiling

The *in silico* prediction of the physicochemical properties and ADME profiling of the test active compounds were performed using the following two web services: SWISSADME and pkCSM web servers. 41,42 In both the web servers, all the data associated with ADME profiling and physicochemical properties of the compounds under investigation were obtained following input of the 2-D molecular structures at the input search bar of the servers.

CRedit authorship contribution statement

Pallavi Saha: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Shashikanta Sau:** Methodology, Formal analysis. **Nitin Pal Kalia:** Supervision, Investigation. **Deepak K. Sharma:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bioorg.2025.108718>.

Data availability

Data will be made available on request.

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